





ACO Access

Product Catalogue and Technical Handbook

Rhinocast® – Ductile Iron Access Covers
Urbanfil® – Galvanised Steel Access Covers
Saku – Thermoplastic Access Covers
Special Application Access Covers

ACO Group

Founded in 1946, the ACO Group manufactures products for the building and construction industry.

ACO was established in Australia in 1993 and has extensive experience manufacturing and supplying access covers, cable enclosures and drainage systems throughout Australia and the world.

ACO Access

ACO manufactures and distributes Australia's most comprehensive range of access covers.

Products are available for every application from architectural to heavy duty industrial.

ACO's manufacturing plant is located at Emu Plains, NSW and supports five offices and warehouses throughout Australia. ACO also supplies products to South East Asia and the Pacific Islands through its partners.

Technical Advice

ACO provides technical advice to ensure the correct product is chosen for the application from concept design to final installation. Product training and professional education seminars provide up to date product information.

Typical applications

- Roads
- Airports
- Ports
- Rail
- Sewer and stormwater
- Tradewaste
- Industrial developments
- Commercial developments
- Educational facilities
- Transport facilities
- Residential buildings
- Internal building areas
- Streetscapes
- Public domains
- Recreational areas
- Car parks



ACO Pty Ltd, Head Office at Emu Plains NSW Australia



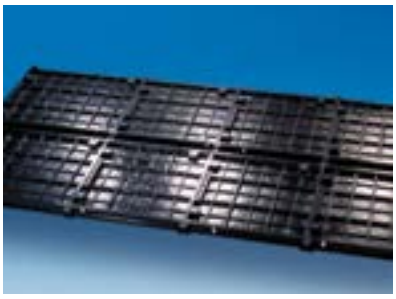
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Rhinocast® – Ductile Iron

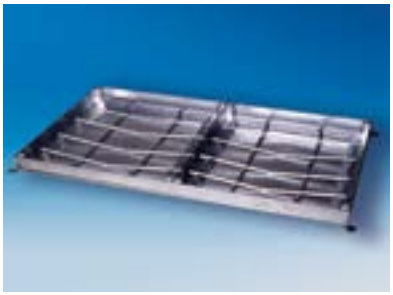
- Circular solid top and recessed
- Square/rectangular solid top and recessed
- two-part, three-part, multi-part and trench runs



Ductile iron multi-part recessed cover

Urbanfil® – Galvanised Steel

- Square/rectangular recessed
- two-part, three-part, multi-part and trench runs
- Pavermate® access covers



Galvanised steel two-part recessed cover

Saku – Thermoplastic

- Circular lightweight solid top



Saku thermoplastic solid top cover

Special Applications

- Servokat - Assisted lift covers
- Fire rated covers





Material information

ACO cover and frame systems are manufactured from either cast iron, galvanised steel or thermoplastic. A choice of materials is available to provide a range of solutions to various application requirements - from loading, health and safety, aesthetics, and lifting to chemical resistance.



Cast Iron

Grey iron is made when pig iron is melted in small cupola furnaces and poured into moulds to make castings. Containing around 2% - 6% carbon, scrap iron or steel is often added to vary the composition.

Ductile iron is produced by adding magnesium to the molten pig iron; when the iron is cast, the carbon forms tiny spherical graphite nodules around the magnesium.

Engineering Properties:

- 600/3 ductile iron to AS 1831:1985
- Tensile Strength - 600 MPa min.
- 0.2% Proof Stress - 370 MPa min.
- Hardness, 145-185 (HRB)
- Density - 7.15 gm/cc
- Elongation - 3% minimum



Galvanised Steel

Hot rolled mild steel is low carbon steel, with traces of manganese, silicon and phosphorus. Recessed access covers fabricated from mild steel are typically reinforced with either 8mm merchant rods or 12mm Tempcore-400Y deformed bars.

Engineering Properties:

- 2mm steel frames and covers to AS 1365
- Hot dip galvanised to AS 4680 for maximum corrosion resistance
- Yield Strength: 280MPa
- Tensile Strength: 395MPa
- Elongation on 80mm %: 35%
- Hardness, 55 (HRB)



Thermoplastic

Thermoplastic resin is a polymer compound that becomes soft or fluid when heated and then returns to its original solid state when cooled.

This compound is typically used in injection moulding whiich requires the raw material to be in liquid or plastic state to fill the mould completely. When cooled, the resin or polymer solidifies to form the finished product.

Engineering Properties:

- Density: 1.13 g/cm^3 @ 23°C
- Tensile strength: 85 MPa @ 50mm/min
- Bending strength: 140 MPa @ 50mm/min
- Strain at Break: 3.5% @ 50mm/min
- Impact Strength: 48 kJ/m^2 @ 23°C
- Offers excellent corrosion and chemical resistance

Product range

ACO cover and frame systems are manufactured from either galvanised steel, thermoplastic or ductile iron. Galvanised steel and thermoplastic access covers are typically used for light duty applications and ductile iron covers for light to extra heavy duty applications.

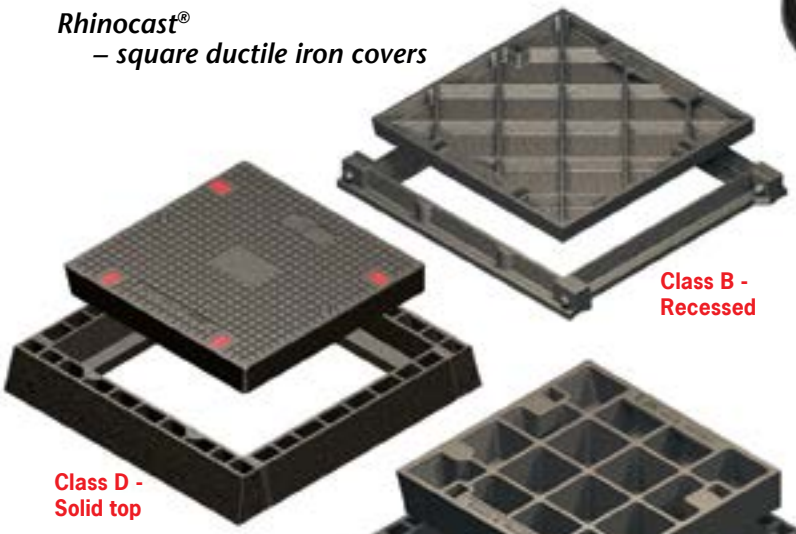


Class B - Solid top

Class D - Recessed

Rhinocast®
- circular ductile iron covers

Rhinocast®
– square ductile iron covers



Class B - Recessed

Class D - Solid top

Class G - Recessed



Class B - Standard

Class B - Pavermate®

Urbanfil®
- galvanised steel covers

Saku
- thermoplastic cover with iron frame

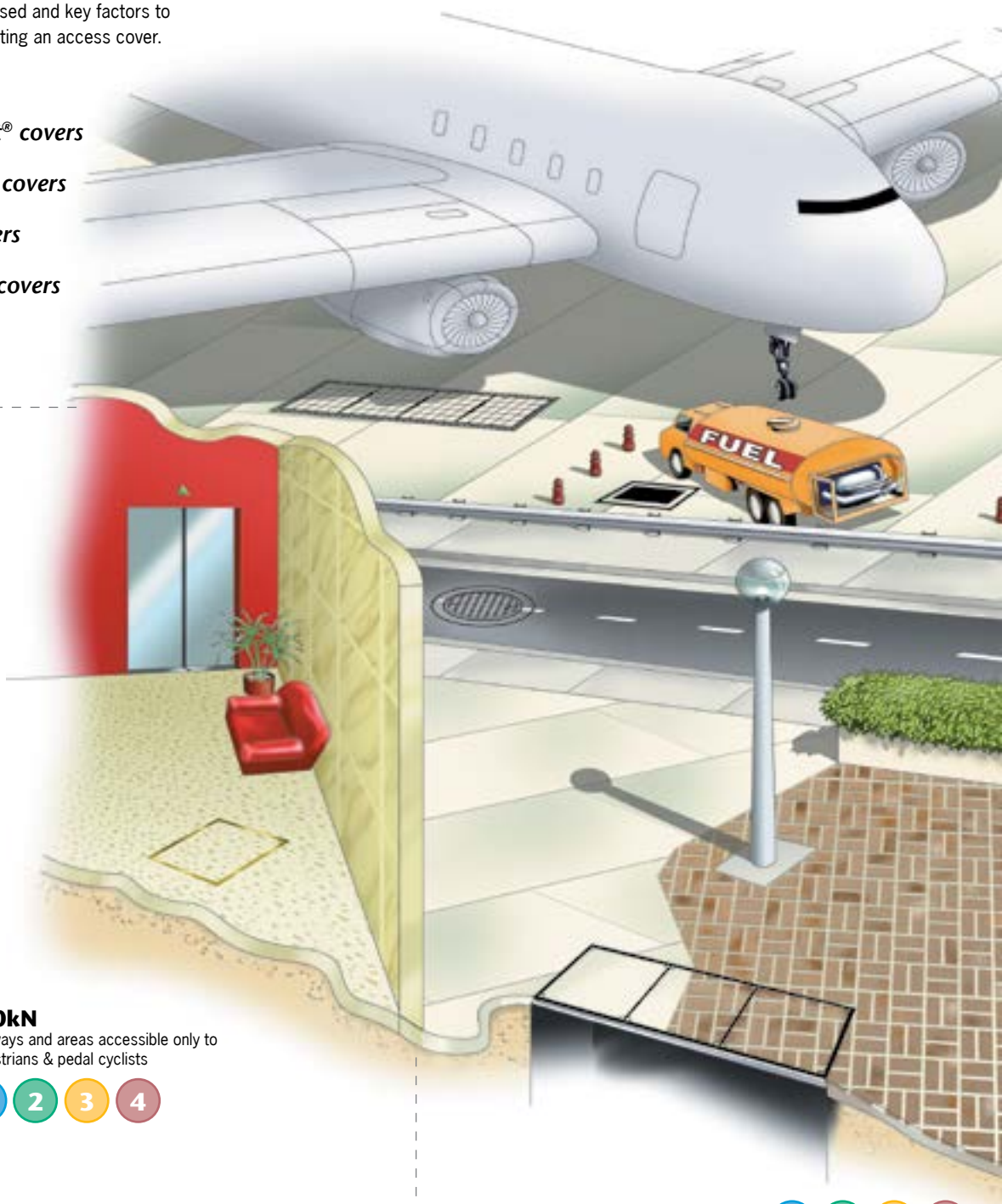


Class B - Solid top

ACO Access product selector

This diagram is provided as a guide to aid selection of the right access cover. It illustrates the various applications where access covers are used and key factors to consider when selecting an access cover.

1. *Rhinocast® covers*
2. *Urbanfil® covers*
3. *Saku covers*
4. *Servokat covers*



G900kN 1 4
Docks and aircraft pavements subject to very high wheel loads

A10kN
Footways and areas accessible only to pedestrians & pedal cyclists

1 2 3 4

B80kN 1 2 3 4
Footways that can be mounted by vehicle or livestock, and light tractor paths

NOTE: *The Load Classes shown above are indicative only. It is the customer's responsibility to determine/verify the anticipated design loads for each application. Engineering advice may be necessary.*



E400kN 1 4
General docks and aircraft pavements

F600kN 1 4
Docks and aircraft pavements subject to high wheel loads

B80kN 1 2 3 4
Footpath and paved areas where vehicles mount accidentally or for light vehicles such as tractors and livestock

D210kN 1
Carriageways of roads and areas open to commercial vehicles

B80kN 1 2 3 4
Footways that can be mounted by vehicle or livestock, and light tractor paths

C150kN 1 4
Malls and pedestrian areas open to slow moving commercial vehicles

Choosing the right access cover

Choosing the right access cover for the given application is essential to prevent problems and product failures in the future. The key factors to consider are listed below and covered in more detail on pages 9-11.

ACO can also offer additional advice and assistance in choosing the right access cover.

1 Load Class



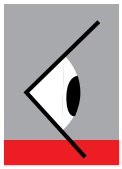
Choose the right cover and frame to withstand the anticipated type and frequency of traffic. ACO covers are designed for AS 3996 loadings. See details on page 9 for full information.

2 Size



Access cover sizes are quoted in terms of clear opening.
ACO offer a range of sizes & formats; Single parts, two-part, Three-part, Trench runs and Multi-parts.

3 Aesthetics



Recessed covers allow matching infill.
Decorative brass or stainless steel edging can be added to rectangular & square covers.

4 Security



Certain covers are bolted as standard, others can have bolts added as an option.

Security Barri bolts are also available.

5 Gas & water tight



All covers are gas and water tight under normal atmospheric pressure.
Hold down bolts are available to assist with back pressure applications.

6 Cover orientation



In fast moving traffic, covers should be positioned with the drawcut facing traffic flow to prevent the cover from accidentally lifting.
Covers should be positioned to avoid obstacles that may prevent easy cover removal.

7 Handling



Smaller individual covers can be combined to create the required opening if there are manual handling restrictions.
ACO also offers a range of thermoplastic covers and a range of assisted lift covers for easier lifting.

1 Load Class



AS 3996 – Clause 1.1 Scope

"This standard specifies requirements for access covers and grates for use in vehicular and pedestrian areas. It applies to access covers & grates having a clear opening of up to 1300mm..."

NATA Certification

As part of ACO's continuous product development and commitment to quality, ACO has NATA certified testing equipment (Licence no. 15193), operated by fully trained and certified technicians

In practice, there are a number of key factors affecting a cover's resistance to load:

i) **Type of traffic** - pedestrians, cars, trucks, forklifts etc. crossing the cover. For trolleys and forklifts particularly consider the weight of loads being carried.

ii) **Frequency of traffic** - more frequent traffic may require a heavier load class.

iii) **Speed of traffic** - fast moving traffic can intensify the load effect on the cover.

iv) **Position of cover** - if the cover is positioned where traffic will be turning, braking or if the cover is installed at the bottom of a ramp, it will be subjected to extreme forces. Selecting the right cover and frame is essential.

v) **Wheel type** - solid tyres exert loads through smaller contact areas than pneumatic tyres. A heavier duty cover may be required.

AS 3996 table of load classification









	Load Class A	Load Class B	Load Class C	Load Class D	Load Class E	Load Class F	Load Class G
	10kN Extra light duty	80kN Light duty	150kN Medium duty	210kN Heavy duty	400kN Extra heavy duty	600kN Extra heavy duty	900kN Extra heavy duty
Typical uses	Footways and areas accessible only to pedestrians & pedal cyclists	Footways that can be mounted by vehicle or livestock, and light tractor paths	Malls and pedestrian areas open to slow moving commercial vehicles	Carriageways of roads and areas open to commercial vehicles	General docks and aircraft pavements	Docks and aircraft pavements subject to high wheel loads	Docks and aircraft pavements subject to very high wheel loads
Approximate wheel load	330kg	2,670kg	5,000kg	8,000kg	13,700kg	20,000kg	30,000kg

EN124 table of load classification

Class A 15kN | Class B 125kN | Class C 250kN | Class D 400kN | Class E 600kN | Class F 900kN

Rhinocast® - Ductile Iron Covers

- Circular covers - p14
- Square/rectangular covers [1] - p16

Urbanfil® - Galvanised Steel Covers

- Standard covers - p28
- Pavermate® cover [1] - p32
- Special sizes made to order - contact ACO for details

Saku - Thermoplastic Covers

- Circular solid covers - p36

Servokat - Assisted Lift Covers [2]

- Square/rectangular covers - p39

Notes:

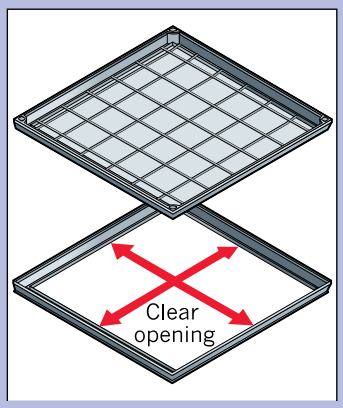
[1] In block paver areas, pavers can decrease the stated loading of the access cover.

[2] Not certified to AS 3996 but have been tested to stated loads under different loading standards (EN 124). The industry standard is for covers to be tested in the single cover format.

2 Size

Clear opening
The unobstructed opening inside the frame. Dimensions are given as width (W) by length (L).

Ductile iron covers are specified with the width parallel to the lifting ends and undercut. Length is parallel to the direction of cover removal.



Single cover
An access cover where a single cover is used.

Two-part
An access cover where two covers are seated lengthways on a single frame.

Three-part
An access cover where three covers are seated lengthways an a single frame.

Trench Run
An access cover where multiple covers are seated lengthways on a single frame.

Multi-part
An access cover where multiple covers are seated both lengthways and widthways on a single frame. Beams are required to support the covers but are removable to provide full access.

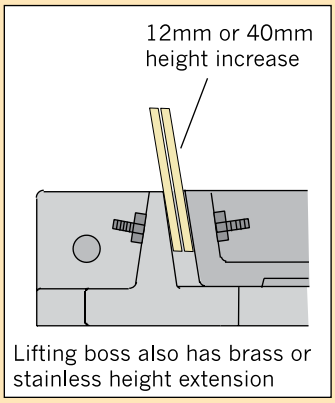
3 Aesthetics

Infill materials
Recessed covers offer the ability to fill the cover with material to match or complement the surrounding pavement.

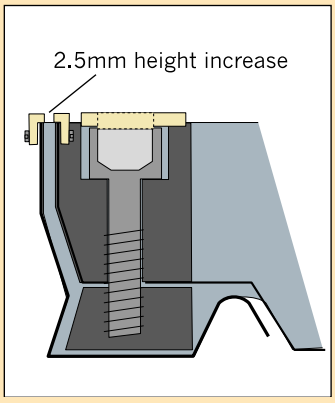
A maximum tile depth of 25mm and maximum paver depth of 40mm is recommended.

Tiles or pavers must be fully restrained and bonded to the concrete bed to prevent damage to the cover. An epoxy mortar is recommended.

Decorative edging
A strip of stainless steel or brass can be fixed to the edge of the cover and frame for an attractive finish.



Ductile iron covers
Height increase : 12 or 40mm
Width/Length increase : 6mm



Galvanised steel covers
Height increase : 2.5mm
Width/Length increase : 2.5mm

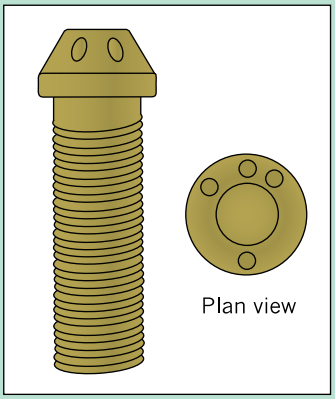
4 Security

Locking
For additional security or for back pressure applications, locking bolts can be added to the cover. Locking bosses are fitted to the cover. The cover and frame is drilled and tapped to accept the locking bolt.

Note: the following covers are locked as standard;

- Ductile iron solid top circular covers
- All Urbanfil® galvanised steel covers
- Saku covers
- Servokat covers

Barri Bolt
A tamper resistant locking bolt for security applications. Special tools are required to remove the bolt.



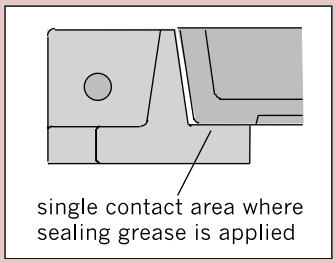
5 Gas and Water seal

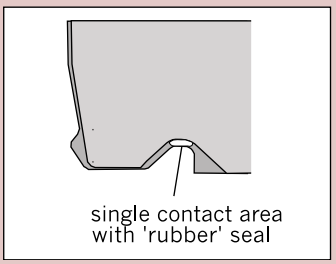


All standard covers are gas and water sealed as standard, to normal atmospheric pressure (up to 1kPa). This type of seal also offers a seal against odours.

Pressure tight
For applications where back pressure is over 1kPa, the addition of locking bolts prevents the ingress of gas or water.

Single Seal
There is one point of contact between the frame and cover where the seal is achieved. The seal can be achieved with grease (Rhinocast®) or a 'rubber' gasket (Urbanfil®, Saku & Servokat).

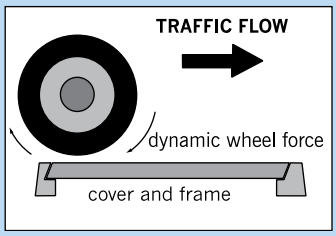




6 Traffic Flow

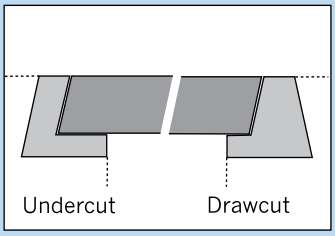


Traffic Flow
For Rhinocast® covers, the drawcut edge should face the orientation traffic flow to prevent cover lifting.



Drawcut
Top of cover is set back from bottom.

Undercut
Top of cover overhangs the bottom.



7 Handling



Work Cover Lifting Guidelines
Work Cover National Code of Practice for Manual Handling recommends a maximum unassisted lifting weight of 55kg. Weights above this require the use of mechanical lifters. All Saku covers comply with this code.

Note: Certain states/companies may have different maximum lifting requirements.

Assisted lift
A gas strut is fitted to the frame and cover to enable the cover to be easily lifted.
Refer to Servokat covers (pg 39).

Lifting Keys
Ductile iron, galvanised steel and thermoplastic covers can be lifted using standard Australian lifting keys. A selection of short handle, long handle and mechanical lifters are available - see pg 42 - 44.

Other Commonly Used Terms

Anti-slip surface
A textured finish on solid top covers to reduce the risk of slipping.

Concrete ties
Profile that holds the frame into the concrete bed and prevents the frame being lifted out of its surround.

Keyhole cap
A cover above the keyhole to prevent dirt and debris ingress.

Lead seal
Lead is used to seal the joint between cast iron frames. Lead provides a flexible seal that does not deteriorate in extreme temperatures.

Lifting boss
The recess where the lifting key is inserted and turned to enable the cover to be lifted. ACO's ductile iron and galvanised steel covers use standard lifting keys to AS 3996.

Recessed cover
A cover that requires a concrete infill material added on site. Also allows infill paving materials to compliment or match surrounding area.

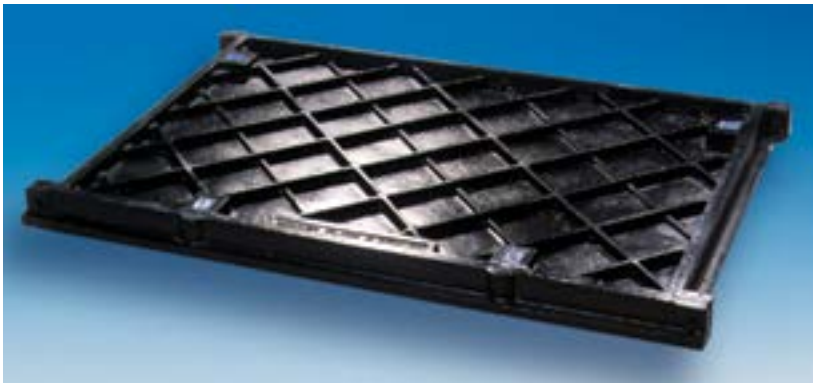
Reo-bar
Steel bars used for reinforcing galvanised steel covers. These are integral to the covers' strength.

Seating
The frame has an angle at the bottom upon which the cover sits and seals.

Rhinocast® – Ductile iron access covers and frames

Typical Applications

- Internal areas
- Streetscapes
- Prestige/Aesthetic projects
- Commercial areas
- Industrial applications
- Docks and ports
- Airports
- Ducting trenches
- Roads and carriageways
- Electrical and communications
- Tradewaste, sewer and stormwater



Ductile iron strengthening ribs cast into the cover.



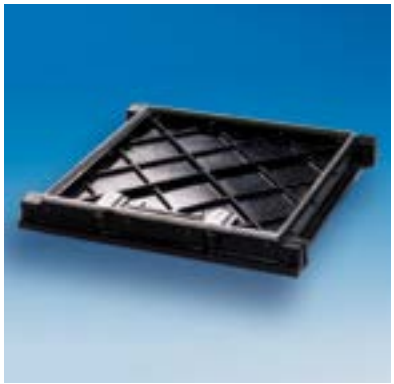


Decorative edge can be attached to the cover and frame for aesthetic areas and adds 12 or 40mm to overall height to allow for use of tiles or pavers

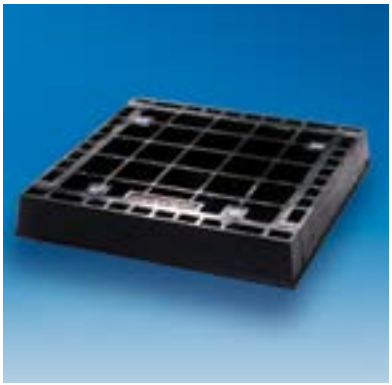


Circular covers are available as recessed or solid top

Class B Cover with 'T' frame



Class D & G covers with cellular frame



Circular covers



Product Features

- All covers are designed to meet Australian Standard AS 3996 Class B, C, D or G loading
- 600/3 ductile iron cover to AS 1831 and grey iron frame
- All covers are sealed to 1.5kPa to AS 3996 requirements
- 'T' or cellular frame provides excellent anchoring of the frame into the concrete bed
- Drawcut/undercut profile of cover and frame allows the cover to slide out, in one direction only
- Reinforcing ribs cast into the cover provide required strength
- Plastic dust caps prevent lifting boss filling with dirt and debris
- Optional brass or stainless steel decorative edging adds 12mm or 40mm to overall height to allow for use with tiles or brick pavers

Certified Product - AS3996
Global-Mark.com.au®
ID Number: 101573



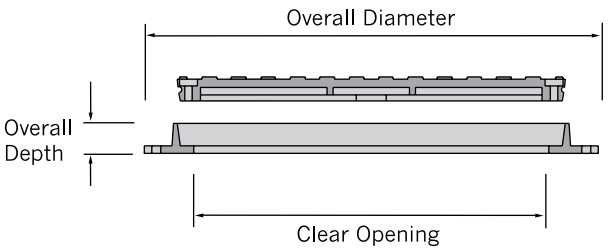


Rhinocast® – Circular solid top ductile iron access cover and frame

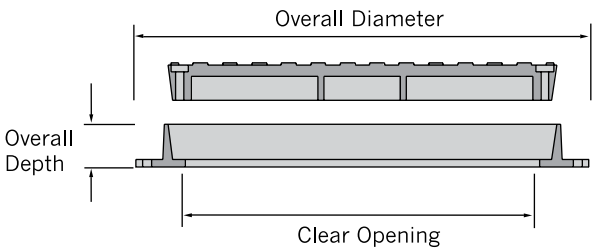
Circular solid top units can be used to provide access to underground stormwater/sewer systems, electricity and communication enclosures.



Class B - 'T' Frame



Class D - 'T' Frame

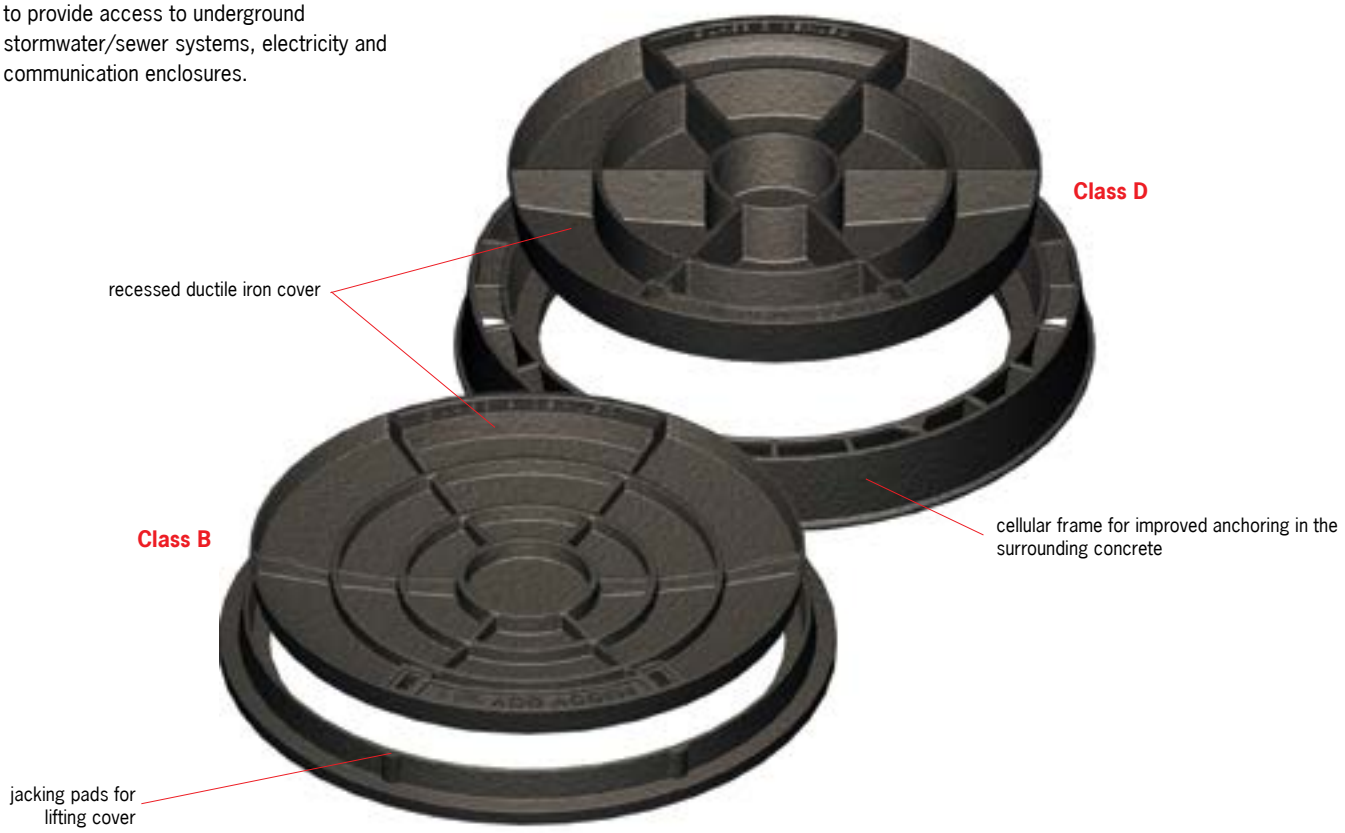


Type	Part No.	Clear Opening Dia (mm)	Overall Dia (mm)	Overall Depth (mm)	Cover Wgt (kg)	Overall Wgt (kg)	ID Label	Locking Option**
Class B - AS 3996 - 80kN								
600SB	**85142**	600	700	50	28.0	43.0	sewer	LD/BB
615SB	**85134**	615	715	55	29.0	48.0	sewer	BB
600SB	**85196**	600	700	50	28.0	43.0	stormwater	LD/BB
Class D - AS 3996 - 210kN								
600SD	**89155**	600	700	72	43.0	63.0	stormwater	LD/BB
615SD	**89148**	615	715	72	45.0	65.0	sewer	BB
600SD	**89183**	600	700	72	43.0	63.0	sewer	LD/BB

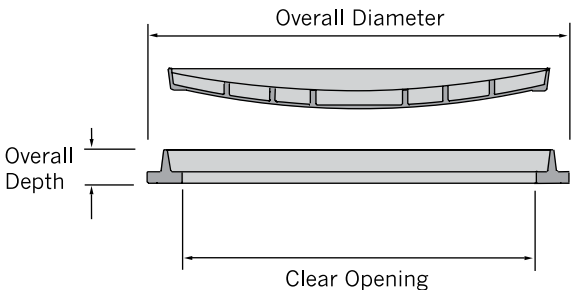
* Decorative edging - not available with circular covers.
** Solid top covers have locking bolts as standard, for security applications, specify Barri bolts (BB) after part no.

Rhinocast® – Circular recessed ductile iron access cover and frame

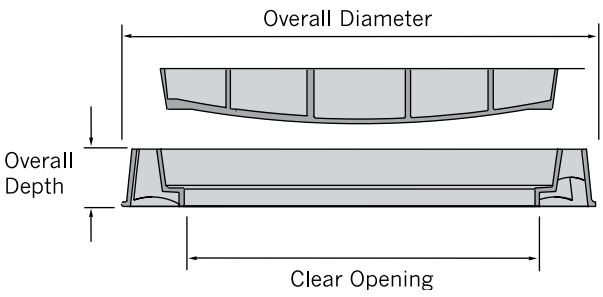
Circular recessed units can be used to provide access to underground stormwater/sewer systems, electricity and communication enclosures.



Class B - 'T' Frame



Class D - Cellular Frame

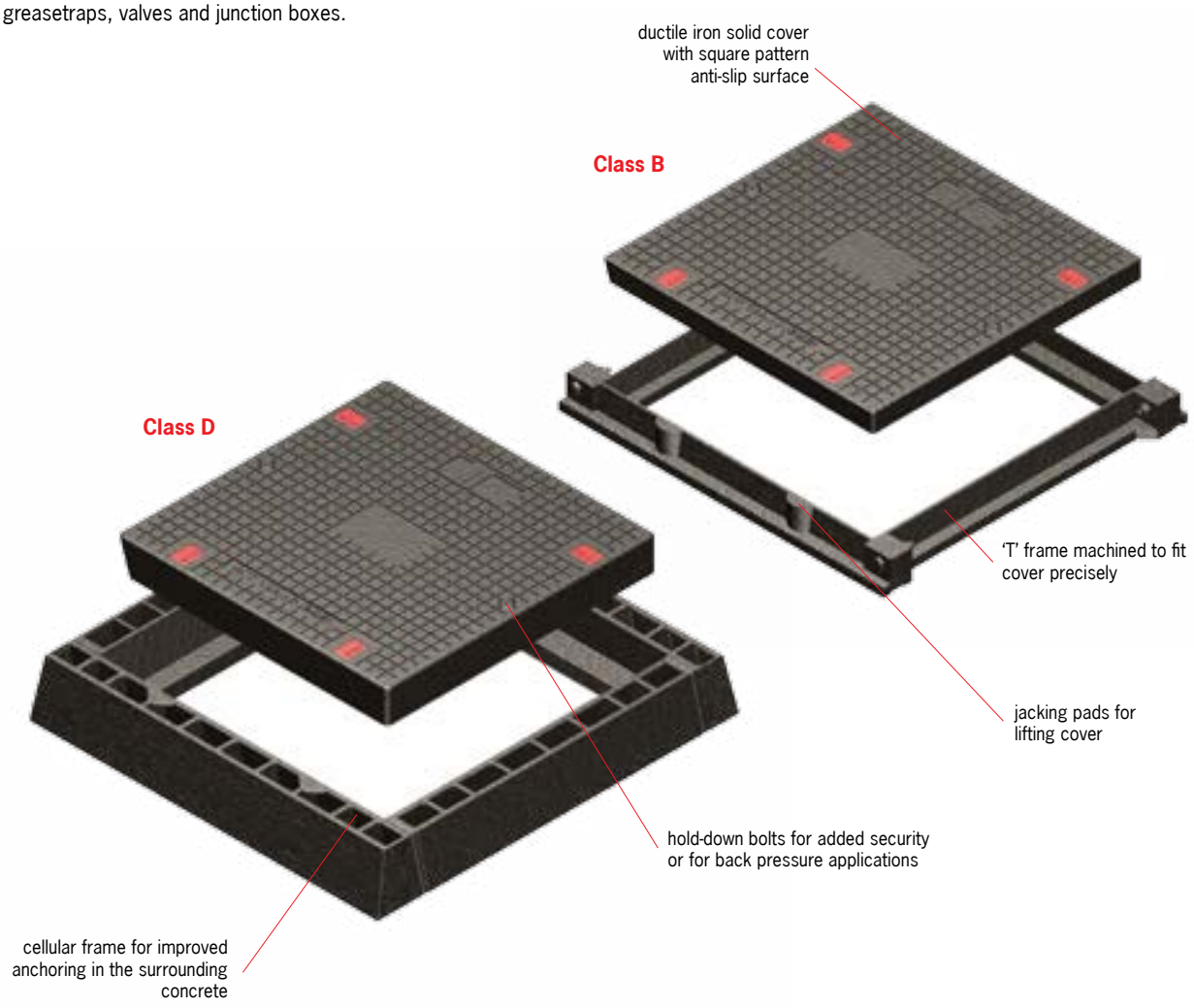


Type	Part No.	Clear Opening Dia (mm)	Overall Dia (mm)	Overall Depth (mm)	Cover Wgt (kg)	Overall Wgt (kg)	Decorative Edging*	Locking Option**
Class B - AS 3996 - 80kN								
600RB	**85107**	600	720	58	30.0	55.0	n/a	LD/BB
Class D - AS 3996 - 210kN								
600RD	**89109**	600	815	102	51.0	98.0	n/a	LD/BB

* Decorative edging - not available with circular covers.
** Optional locking bolts are available for security or back pressure applications - specify standard bolts (LD) or Barri bolts (BB) after part no.

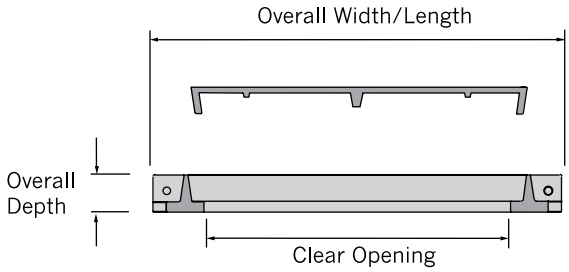
Rhinocast® – Single part solid top ductile iron access cover and frame

Single part square units can be used to provide access to underground stormwater and ewer systems, electricity and communication enclosures, greasetraps, valves and junction boxes.

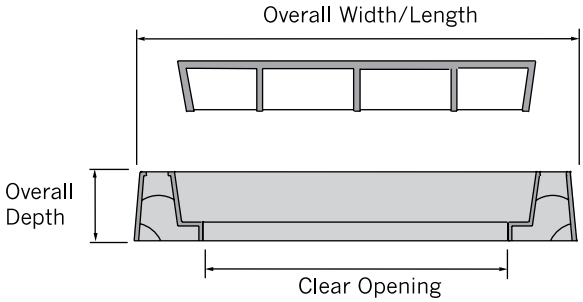


Also available as 2 part,3 part, trench and multi-part systems

Class B - 'T' Frame



Class D/G - Cellular Frame



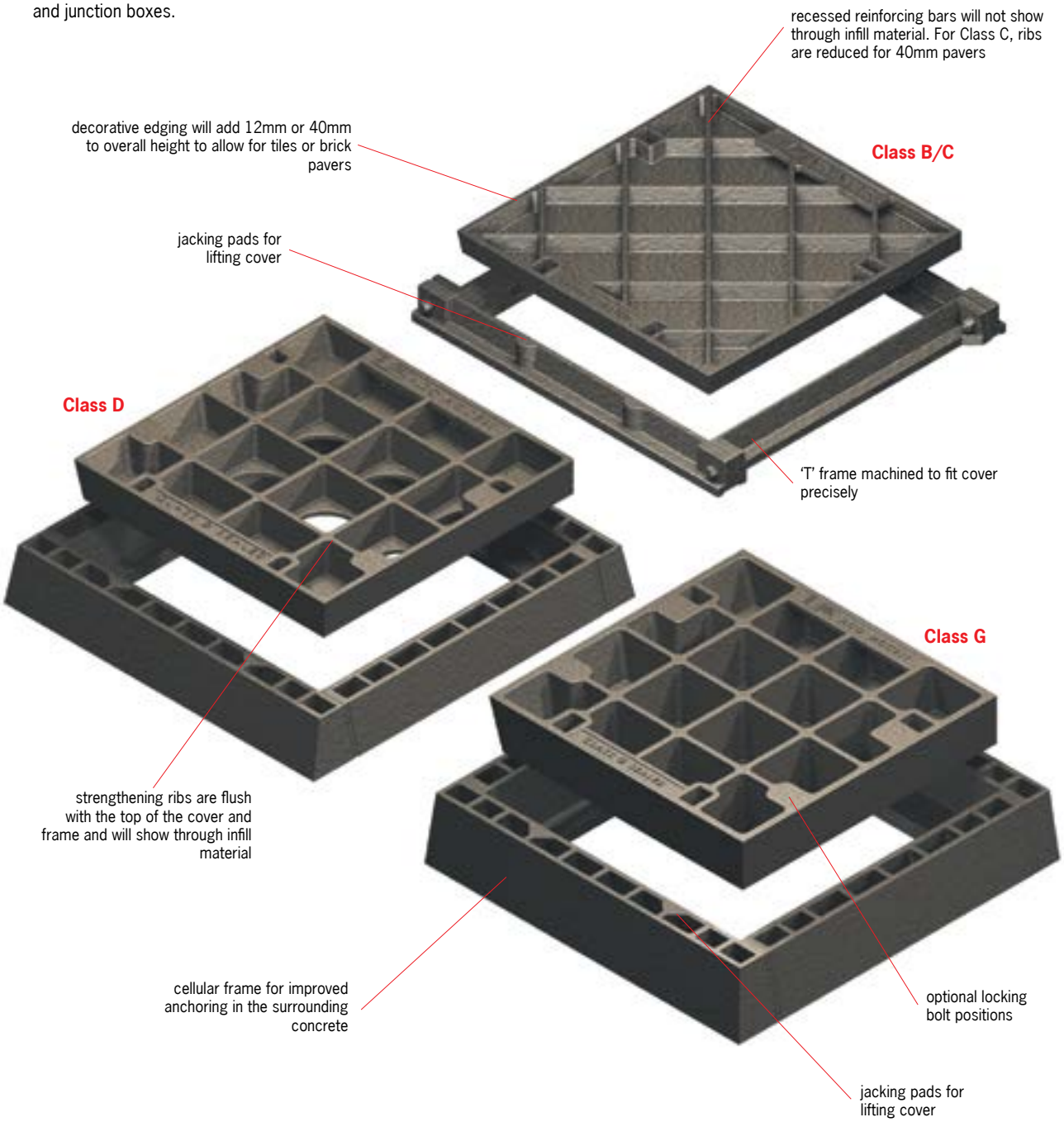
Type	Part No.	Clear Opening* W x L (mm)	Overall W x L (mm)	Overall Depth (mm)	Cover Wgt (kg)	Overall Wgt (kg)	Decorative Edging*	Locking Option**
Class B - AS 3996 - 80kN								
DIS44B	**85249**	450 x 450	600 x 560	55	20.0	44.0	BE/SS - 12/40	LD/BB
DIS46B	**85261**	450 x 600	600 x 710	55	31.0	54.0	BE/SS - 12/40	LD/BB
DIS64B	**85236**	600 x 450	750 x 560	55	28.0	54.0	BE/SS - 12/40	LD/BB
DIS66B	**85222**	600 x 600	750 x 710	55	35.0	62.0	BE/SS - 12/40	LD/BB
DIS67B	**85215**	600 x 750	750 x 860	55	51.0	82.0	BE/SS - 12/40	LD/BB
DIS74B	**85298**	750 x 450	900 x 560	55	39.0	66.0	BE/SS - 12/40	LD/BB
DIS76B	**85287**	750 x 600	900 x 710	55	52.0	82.0	BE/SS - 12/40	LD/BB
DIS77B	**85274**	750 x 750	900 x 860	55	60.0	94.0	BE/SS - 12/40	LD/BB
DIS94B	**85208**	900 x 450	1050 x 560	55	48.0	79.0	BE/SS - 12/40	LD/BB
DIS96B	**85253**	900 x 600	1050 x 710	55	57.0	89.0	BE/SS - 12/40	LD/BB
Class D - AS 3996 - 210kN								
DIS44D	**89268**	450 x 450	660 x 660	100	41.0	87.0	BE/SS - 12/40	LD/BB
DIS46D	**89283**	450 x 600	660 x 810	100	59.0	101.0	BE/SS - 12/40	LD/BB
DIS64D	**89251**	600 x 450	810 x 660	100	59.0	101.0	BE/SS - 12/40	LD/BB
DIS66D	**89235**	600 x 600	810 x 810	100	67.0	122.0	BE/SS - 12/40	LD/BB
DIS67D	**89227**	600 x 750	810 x 960	100	81.0	142.0	BE/SS - 12/40	LD/BB
DIS74D	**89294**	750 x 450	960 x 660	100	81.0	120.0	BE/SS - 12/40	LD/BB
DIS76D	**89309**	750 x 600	960 x 810	100	94.0	140.0	BE/SS - 12/40	LD/BB
DIS77D	**89315**	750 x 750	960 x 960	100	75.0	161.0	BE/SS - 12/40	LD/BB
DIS94D	**89215**	900 x 450	1110 x 660	100	98.0	142.0	BE/SS - 12/40	LD/BB
DIS96D	**89276**	900 x 600	1110 x 810	100	106.0	162.0	BE/SS - 12/40	LD/BB

* Decorative Brass (BE) or Stainless Steel (SS) edging can be added - just specify BE or SS and height (12mm or 40mm) after part no.

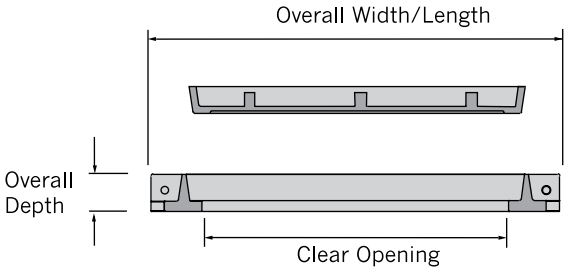
** Optional locking bolts are available for security or back pressure applications - specify standard bolts (LD) or Barri bolts (BB) after part no.

Rhinocast® – Single part recessed ductile iron access cover and frame

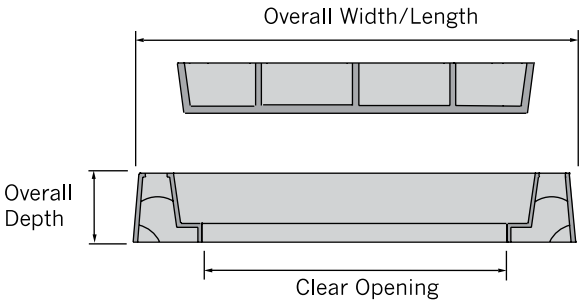
Single part square units can be used to provide access to underground stormwater/sewer systems, electricity and communication enclosures, valves and junction boxes.



Class B - 'T' Frame



Class D/G - Cellular Frame



Type	Part No.	Clear Opening* W x L (mm)	Overall W x L (mm)	Overall Depth (mm)	Cover Wgt (kg)	Overall Wgt (kg)	Decorative Edging*	Locking Option**
Class B - AS 3996 - 80kN								
DI44B	**85014**	450 x 450	600 x 560	55	28.0	47.0	BE/SS - 12/40	LD/BB
DI46B	**85020**	450 x 600	600 x 710	55	30.0	56.0	BE/SS - 12/40	LD/BB
DI64B	**85037**	600 x 450	750 x 560	55	28.0	54.0	BE/SS - 12/40	LD/BB
DI66B	**85045**	600 x 600	750 x 710	55	36.0	63.0	BE/SS - 12/40	LD/BB
DI67B	**85125**	600 x 750	750 x 860	55	43.0	74.0	BE/SS - 12/40	LD/BB
DI76B	**85052**	750 x 600	900 x 710	55	44.0	74.0	BE/SS - 12/40	LD/BB
DI77B	**85061**	750 x 750	900 x 860	55	60.0	94.0	BE/SS - 12/40	LD/BB
DI94B	**85119**	900 x 450	1050 x 560	55	45.0	76.0	BE/SS - 12/40	LD/BB
DI96B	**85078**	900 x 600	1050 x 710	55	57.0	89.0	BE/SS - 12/40	LD/BB
DI97B	**85086**	900 x 750	1050 x 860	55	71.0	107.0	BE/SS - 12/40	LD/BB
DI99B	**85093**	900 x 900	1050 x 1010	55	86.0	124.0	BE/SS - 12/40	LD/BB
Class C - AS 3996 - 150kN - reduced ribs for 40mm deep pavers								
DI44C	**142313**	450 x 450	630 x 619	85	30.0	63.0	BE/SS - 12/40	LD/BB
DI66C	**142314**	600 x 600	780 x 769	85	56.0	98.0	BE/SS - 12/40	LD/BB
DI96C	**142315**	900 x 600	1080 x 769	85	75.0	123.0	BE/SS - 12/40	LD/BB
DI99C	**142316**	900 x 900	1080 x 1069	85	102.0	160.0	BE/SS - 12/40	LD/BB
Class D - AS 3996 - 210kN								
DI44D	**89013**	450 x 450	665 x 665	100	42.0	85.0	BE/SS - 12/40	LD/BB
DI46D	**89020**	450 x 600	665 x 815	100	51.0	100.0	BE/SS - 12/40	LD/BB
DI64D	**89034**	600 x 450	815 x 665	100	51.0	100.0	BE/SS - 12/40	LD/BB
DI66D	**89041**	600 x 600	815 x 815	100	62.0	116.0	BE/SS - 12/40	LD/BB
DI67D	**89137**	600 x 750	815 x 965	100	74.0	134.0	BE/SS - 12/40	LD/BB
DI76D	**89058**	750 x 600	965 x 815	100	77.0	136.0	BE/SS - 12/40	LD/BB
DI77D	**89065**	750 x 750	965 x 965	100	92.0	157.0	BE/SS - 12/40	LD/BB
DI94D	**89124**	900 x 450	1115 x 665	100	72.0	130.0	BE/SS - 12/40	LD/BB
DI96D	**89072**	900 x 600	1115 x 815	100	88.0	151.0	BE/SS - 12/40	LD/BB
DI97D	**89086**	900 x 750	1115 x 965	100	106.0	175.0	BE/SS - 12/40	LD/BB
DI99D	**89093**	900 x 900	1115 x 1115	100	136.0	209.0	BE/SS - 12/40	LD/BB
Class G - AS 3996 - 900kN								
DI44G	**87016**	450 x 450	685 x 685	125	86.0	149.0	BE/SS - 12/40	LD/BB
DI64G	**87039**	600 x 450	835 x 685	125	104.0	170.0	BE/SS - 12/40	LD/BB
DI66G	**87044**	600 x 600	835 x 835	125	135.0	206.0	BE/SS - 12/40	LD/BB
DI77G	**87067**	750 x 750	985 x 985	125	200.0	251.0	BE/SS - 12/40	LD/BB
DI94G	**87082**	900 x 450	1135 x 685	125	145.0	230.0	BE/SS - 12/40	LD/BB
DI96G	**87075**	900 x 600	1135 x 835	125	180.0	270.0	BE/SS - 12/40	LD/BB

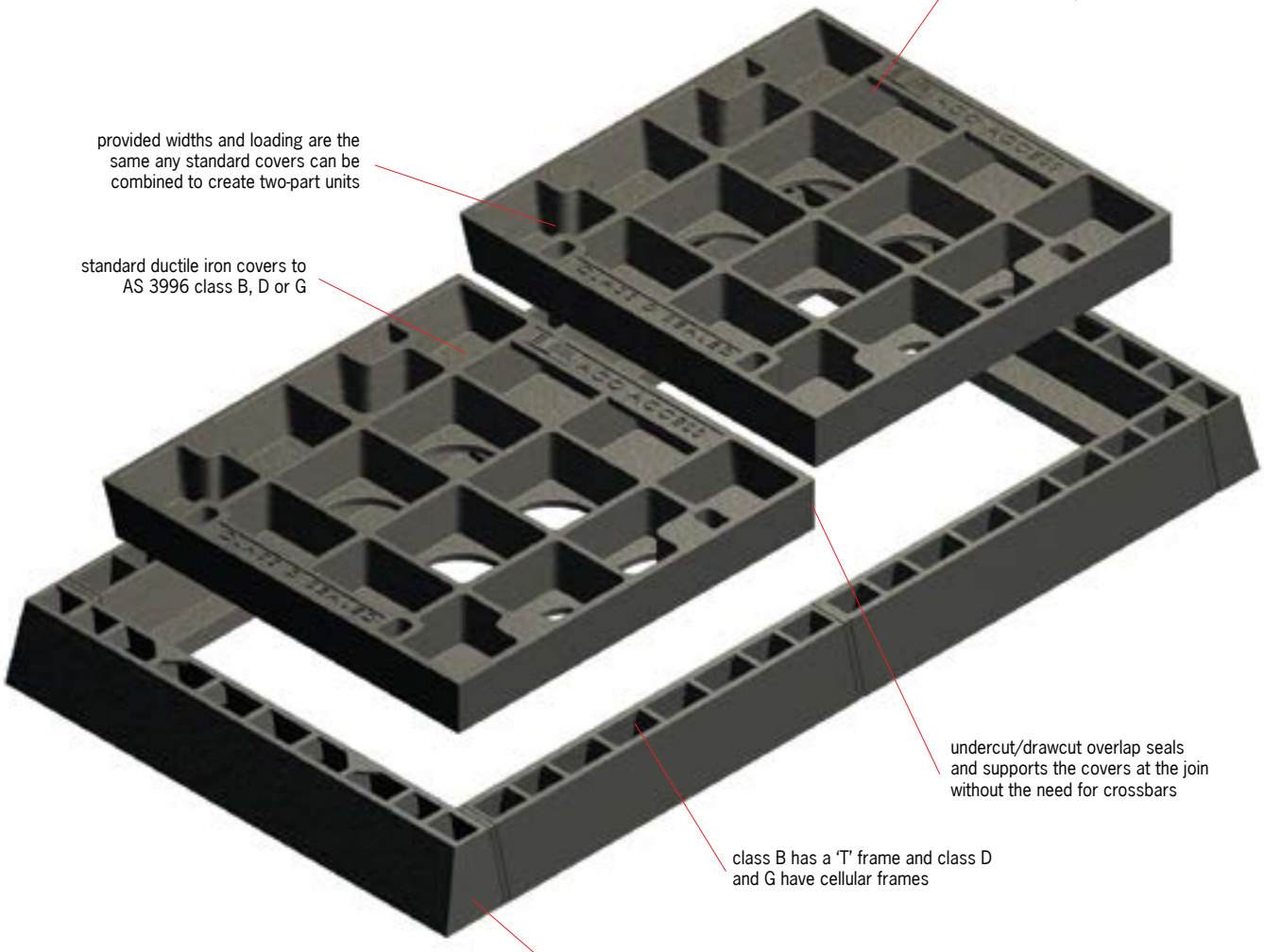
* Decorative Brass (BE) or Stainless Steel (SS) edging can be added - just specify BE or SS and height (12mm or 40mm) after part no.

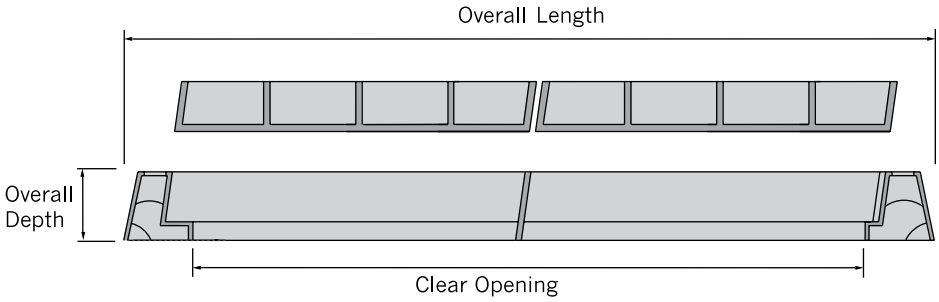
** Optional locking bolts are available for security or back pressure applications - specify standard bolts (LD) or Barri bolts (BB) after part no.

Rhinocast® – Two-part recessed ductile iron access cover and frame

Two-part units combine two standard size covers in a single run. They are used where a single part cover does not offer a large enough clear opening or if smaller individual covers are required for easier lifting.

Common applications include grease traps/arrestors, access to underground stormwater/sewer systems, electricity and communication enclosures, valves and junction boxes.





Type	Part No.	Clear Opening* W x L (mm)	Overall W x L (mm)	Overall Depth (mm)	Cover Wgt (kg)	Overall Wgt (kg)	Decorative Edging*	Locking Option**
	Class B - AS 3996 - 80kN							
DI249B	**85418**	450 x 945	600 x 1055	55	22.0/22.0	83.0	BE/SS - 12/40	LD/BB
DI2412B	**85433**	450 x 1245	600 x 1355	55	30.0/30.0	100.0	BE/SS - 12/40	LD/BB
DI269B	**85444**	600 x 945	750 x 1055	55	28.0/28.0	98.0	BE/SS - 12/40	LD/BB
DI2610B	**85459**	600 x 1095	750 x 1205	55	28.0/36.0	105.0	BE/SS - 12/40	LD/BB
DI2612B	**85465**	600 x 1245	750 x 1355	55	36.0/36.0	110.0	BE/SS - 12/40	LD/BB
DI2613B	**85426**	600 x 1395	750 x 1505	55	36.0/43.0	123.0	BE/SS - 12/40	LD/BB
DI2615B	**85555**	600 x 1545	750 x 1655	55	43.0/43.0	132.0	BE/SS - 12/40	LD/BB
DI2712B	**85477**	750 x 1245	900 x 1355	55	44.0/44.0	132.0	BE/SS - 12/40	LD/BB
DI2713B	**85482**	750 x 1395	900 x 1505	55	44.0/60.0	152.0	BE/SS - 12/40	LD/BB
DI2715B	**85499**	750 x 1545	900 x 1655	55	60.0/60.0	172.0	BE/SS - 12/40	LD/BB
DI299B	**85159**	900 x 945	1050 x 1055	55	45.0/45.0	142.0	BE/SS - 12/40	LD/BB
DI2910B	**85510**	900 x 1095	1050 x 1205	55	45.0/57.0	153.0	BE/SS - 12/40	LD/BB
DI2912B	**85503**	900 x 1245	1050 x 1355	55	57.0/57.0	166.0	BE/SS - 12/40	LD/BB
DI2915B	**85524**	900 x 1545	1050 x 1655	55	71.0/71.0	198.0	BE/SS - 12/40	LD/BB
DI2916B	**85531**	900 x 1695	1050 x 1805	55	71.0/86.0	213.0	BE/SS - 12/40	LD/BB
	Class D - AS 3996 - 210kN							
DI249D	**89415**	450 x 970	665 x 1185	100	42.0/42.0	148.0	BE/SS - 12/40	LD/BB
DI2412D	**89431**	450 x 1270	665 x 1485	100	51.0/51.0	178.0	BE/SS - 12/40	LD/BB
DI269D	**89448**	600 x 970	815 x 1185	100	51.0/51.0	173.0	BE/SS - 12/40	LD/BB
DI2611D	**89456**	600 x 1120	815 x 1335	100	51.0/62.0	189.0	BE/SS - 12/40	LD/BB
DI2612D	**89463**	600 x 1270	815 x 1485	100	62.0/62.0	205.0	BE/SS - 12/40	LD/BB
DI2614D	**89542**	600 x 1420	815 x 1635	100	62.0/74.0	223.0	BE/SS - 12/40	LD/BB
DI2615D	**89556**	600 x 1570	815 x 1785	100	74.0/74.0	241.0	BE/SS - 12/40	LD/BB
DI2712D	**89474**	750 x 1270	965 x 1485	100	77.0/77.0	237.0	BE/SS - 12/40	LD/BB
DI2714D	**89489**	750 x 1420	965 x 1635	100	77.0/92.0	261.0	BE/SS - 12/40	LD/BB
DI2715D	**89495**	750 x 1570	965 x 1785	100	92.0/92.0	282.0	BE/SS - 12/40	LD/BB
DI299D	**89162**	900 x 970	1115 x 1185	100	72.0/72.0	224.0	BE/SS - 12/40	LD/BB
DI2911D	**89422**	900 x 1120	1115 x 1335	100	72.0/88.0	245.0	BE/SS - 12/40	LD/BB
DI2912D	**89507**	900 x 1270	1115 x 1485	100	88.0/88.0	266.0	BE/SS - 12/40	LD/BB
DI2914D	**89512**	900 x 1420	1115 x 1635	100	88.0/106.0	290.0	BE/SS - 12/40	LD/BB
DI2915D	**89529**	900 x 1570	1115 x 1785	100	106.0/106.0	314.0	BE/SS - 12/40	LD/BB
DI2918D	**89533**	900 x 1870	1115 x 2085	100	136.0/136.0	382.0	BE/SS - 12/40	LD/BB
	Class G - AS 3996 - 900kN							
DI249G	**87415**	450 x 990	685 x 1225	125	86.0/86.0	266.0	BE/SS - 12/40	LD/BB
DI649G	**87448**	600 x 990	835 x 1225	125	104.0/104.0	305.0	BE/SS - 12/40	LD/BB
DI2611G	**87456**	600 x 1140	835 x 1375	125	104.0/135.0	341.0	BE/SS - 12/40	LD/BB
DI2612G	**87463**	600 x 1290	835 x 1525	125	135.0/135.0	377.0	BE/SS - 12/40	LD/BB
DI2715G	**87450**	750 x 1590	985 x 1825	125	200.0/200.0	553.0	BE/SS - 12/40	LD/BB
DI299G	**87568**	900 x 990	1135 x 1225	125	145.0/145.0	433.0	BE/SS - 12/40	LD/BB
DI2911G	**87507**	900 x 1140	1135 x 1385	125	145.0/180.0	446.0	BE/SS - 12/40	LD/BB
DI2912G	**87512**	900 x 1290	1135 x 1525	125	180.0/180.0	508.0	BE/SS - 12/40	LD/BB

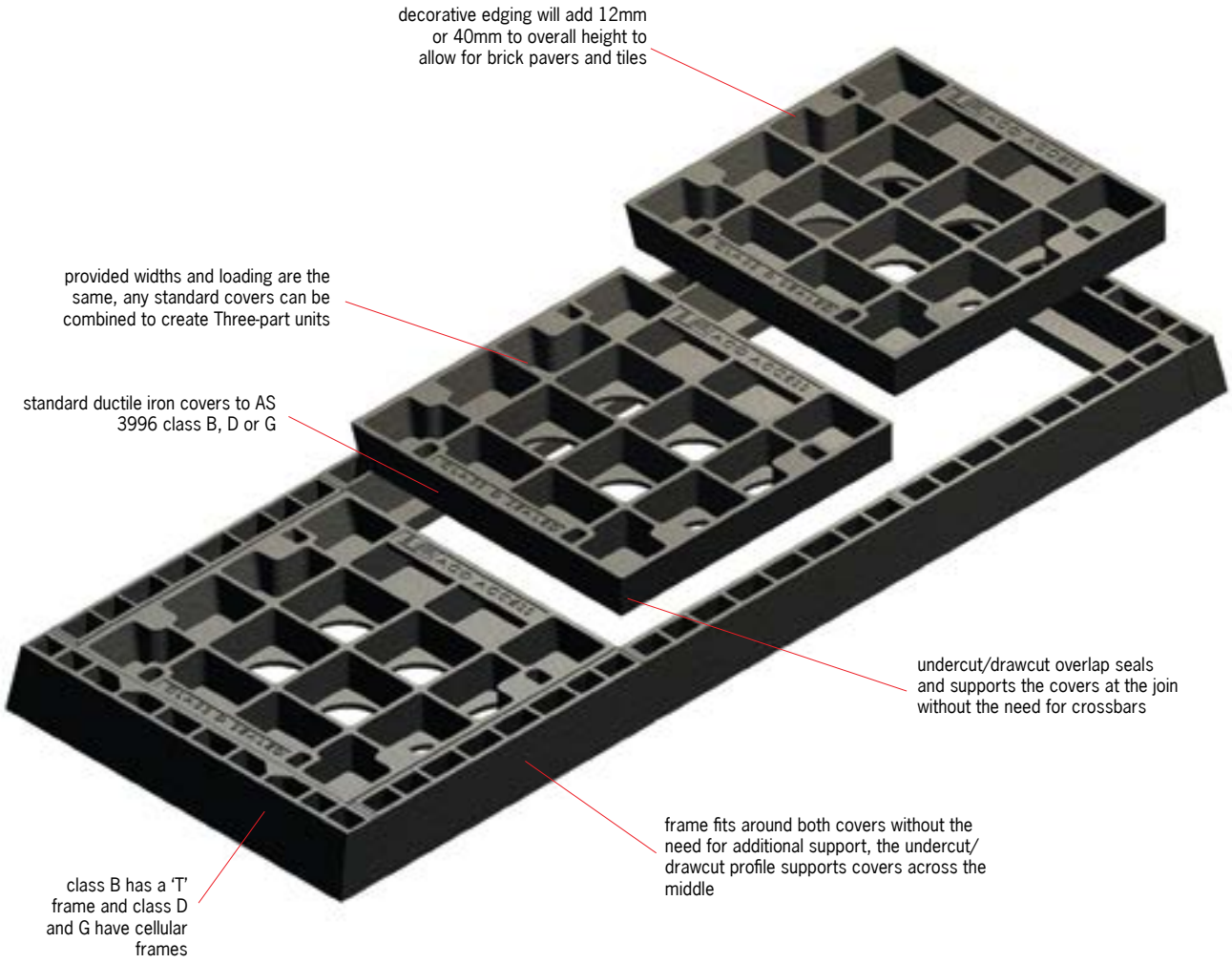
* Decorative Brass (BE) or Stainless Steel (SS) edging can be added - just specify BE or SS and height (12mm or 40mm) after part no.
** Optional locking bolts are available for security or back pressure applications - specify standard bolts (LD) or Barri bolts (BB) after part no.

Rhinocast® – Three-part and trench run recessed ductile iron access covers and frames

Three-part and trench cover units combine standard size covers in a single run. They are used where a single part cover does not offer a large enough clear opening or where continuous access is required.

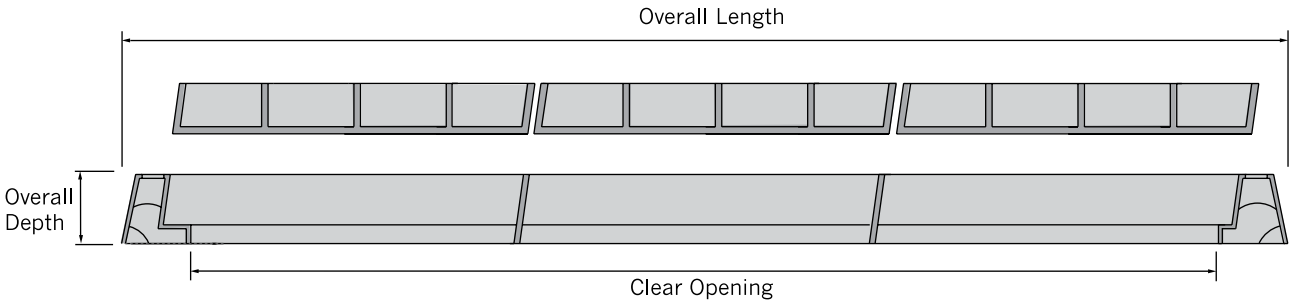
Common applications include grease traps/arrestors, communications enclosures, where access along the entire run is needed. All covers can be removed for full uninterrupted access or single covers can provide access to specific areas.

Covers are available to suit 550, 1,000 and 2,000 litre grease traps.



Trench Runs

Trench runs can be produced using any standard covers in widths of 450, 600, 750 or 900mm and in any length from 2340mm/2390mm upwards in multiples of 150mm.



Type	Part No.	Clear Opening* W x L (mm)	Overall W x L (mm)	Overall Depth (mm)	Cover Wgt (kg)	Overall Wgt (kg)	Decorative Edging*	Locking Option**
Class B - AS 3996 - 80kN								
DI3414B	**85614**	450 x 1440	600 x 1550	55	22/22/22	119.0	BE/SS - 12/40	LD/BB
DI3614B	**85645**	600 x 1440	750 x 1550	55	28/28/28	140.0	BE/SS - 12/40	LD/BB
DI3615B	**85740**	600 x 1590	750 x 1700	55	28/28/36	126.0	BE/SS - 12/40	LD/BB
DI3617B	**85652**	600 x 1740	750 x 1850	55	36/36/28	157.0	BE/SS - 12/40	LD/BB
DI3618B	**85666**	600 x 1890	750 x 2000	55	36/36/36	165.0	BE/SS - 12/40	LD/BB
DI3620B	**85758**	600 x 2040	750 x 2150	55	36/36/43	172.0	BE/SS - 12/40	LD/BB
DI3621B	**85627**	600 x 2190	750 x 2300	55	36/43/43	187.0	BE/SS - 12/40	LD/BB
DI3623B	**85638**	600 x 2340	750 x 2450	55	43/43/43	190.0	BE/SS - 12/40	LD/BB
DI3723B	**85700**	750 x 2340	900 x 2450	55	60/60/60	250.0	BE/SS - 12/40	LD/BB
DI3914B	**85673**	900 x 1440	1050 x 1550	55	45/45/45	206.0	BE/SS - 12/40	LD/BB
DI3915B	**85689**	900 x 1590	1050 x 1700	55	45/45/57	218.0	BE/SS - 12/40	LD/BB
DI3917B	**85717**	900 x 1740	1050 x 1850	55	45/57/57	231.0	BE/SS - 12/40	LD/BB
DI3918B	**85725**	900 x 1890	1050 x 2000	55	57/57/57	243.0	BE/SS - 12/40	LD/BB
DI3923B	**85732**	900 x 2340	1050 x 2450	55	71/71/71	289.0	BE/SS - 12/40	LD/BB
Class D - AS 3996 - 210kN								
DI3414D	**89618**	450 x 1490	665 x 1705	100	42/42/42	219.0	BE/SS - 12/40	LD/BB
DI3614D	**89653**	600 x 1490	815 x 1705	100	51/51/51	252.0	BE/SS - 12/40	LD/BB
DI3616D	**89665**	600 x 1640	815 x 1855	100	51/51/62	283.0	BE/SS - 12/40	LD/BB
DI3617D	**89674**	600 x 1790	815 x 2005	100	51/62/62	299.0	BE/SS - 12/40	LD/BB
DI3619D	**89680**	600 x 1940	815 x 2155	100	62/62/62	300.0	BE/SS - 12/40	LD/BB
DI3620D	**89625**	600 x 2090	815 x 2305	100	62/62/74	322.0	BE/SS - 12/40	LD/BB
DI3622D	**89632**	600 x 2240	815 x 2455	100	62/74/74	330.0	BE/SS - 12/40	LD/BB
DI3623D	**89646**	600 x 2390	815 x 2605	100	74/74/74	342.0	BE/SS - 12/40	LD/BB
DI3723D	**89712**	750 x 2390	965 x 2605	100	92/92/92	411.0	BE/SS - 12/40	LD/BB
DI3914D	**89720**	900 x 1490	1115 x 1705	100	72/72/72	352.0	BE/SS - 12/40	LD/BB
DI3916D	**89738**	900 x 1640	1115 x 1855	100	72/72/88	368.0	BE/SS - 12/40	LD/BB
DI3917D	**89746**	900 x 1790	1115 x 2005	100	72/88/88	394.0	BE/SS - 12/40	LD/BB
DI3919D	**89753**	900 x 1940	1115 x 2155	100	88/88/88	405.0	BE/SS - 12/40	LD/BB
DI3923D	**89764**	900 x 2390	1115 x 2605	100	106/106/106	465.0	BE/SS - 12/40	LD/BB
Class G - AS 3996 - 900kN								
DI3414G	**87613**	450 x 1530	665 x 1765	125	86/86/86	393.0	BE/SS - 12/40	LD/BB
DI3614G	**87658**	600 x 1530	815 x 1765	125	104/104/104	456.0	BE/SS - 12/40	LD/BB
DI3616G	**87665**	600 x 1680	815 x 1915	125	104/104/135	505.0	BE/SS - 12/40	LD/BB
DI3617G	**87672**	600 x 1830	815 x 2065	125	104/135/135	523.0	BE/SS - 12/40	LD/BB
DI3619G	**87686**	600 x 1980	815 x 2215	125	135/135/135	554.0	BE/SS - 12/40	LD/BB
DI3623G	**87693**	750 x 2430	965 x 2665	125	200/200/200	807.0	BE/SS - 12/40	LD/BB
DI3914G	**87709**	900 x 1530	1115 x 1765	125	145/145/145	636.0	BE/SS - 12/40	LD/BB
DI3916G	**87714**	900 x 1640	1115 x 1915	125	145/145/180	671.0	BE/SS - 12/40	LD/BB
DI3917G	**87720**	900 x 1830	1115 x 2065	125	145/180/180	711.0	BE/SS - 12/40	LD/BB
DI3919G	**87737**	900 x 1980	1115 x 2215	125	180/180/180	746.0	BE/SS - 12/40	LD/BB

* Decorative Brass (BE) or Stainless Steel (SS) edging can be added - just specify BE or SS and height (12mm or 40mm) after part no.
** Optional locking bolts are available for security or back pressure applications - specify standard bolts (LD) or Barri bolts (BB) after part no.

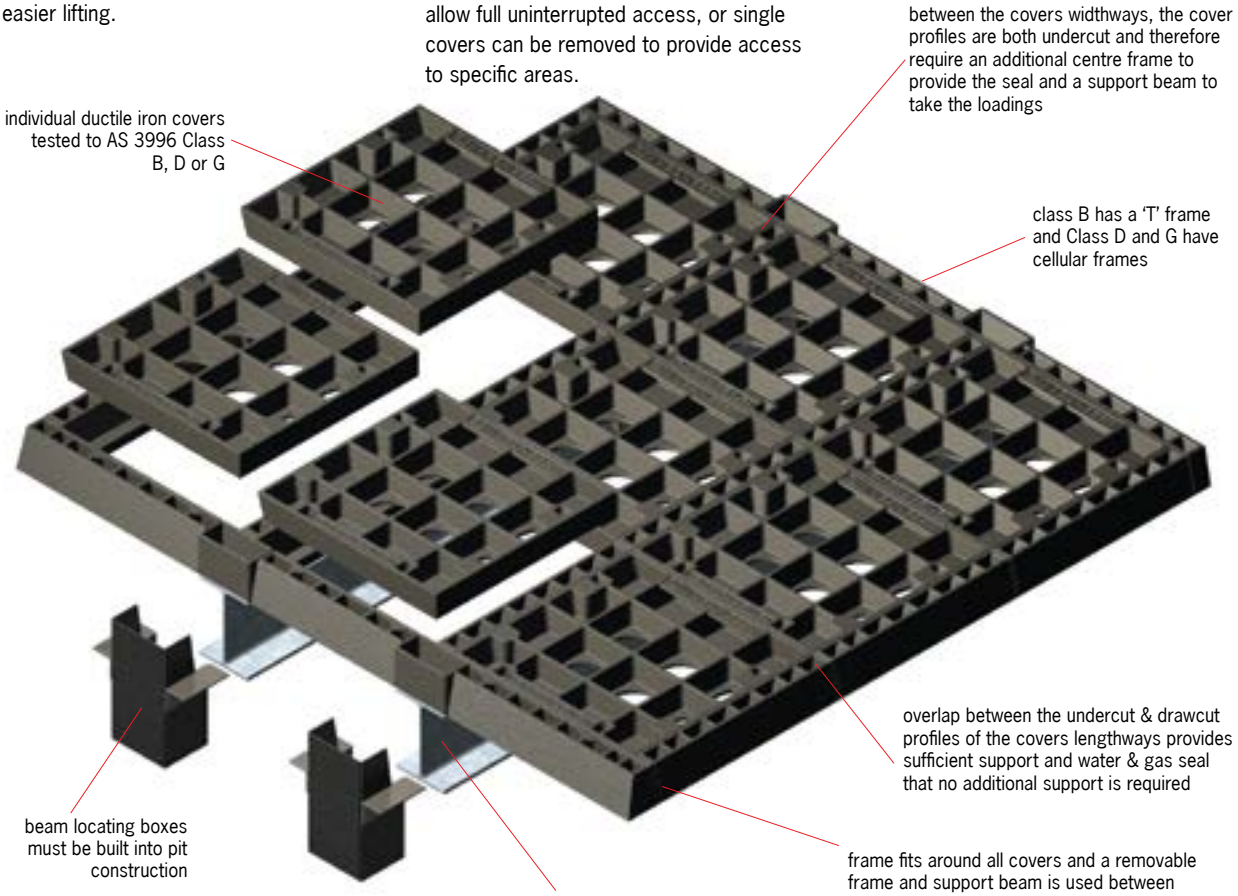
Rhinocast® – Multi-part recessed ductile iron access covers and frame

Multi-part units combine standard size covers in multiple rows and lengths. They are used where a single part cover does not offer a large enough clear opening or if smaller individual covers are required for easier lifting.

Covers do not have to be the same length, provided each row has covers of the same width and load class.

All covers and beams can be removed to allow full uninterrupted access, or single covers can be removed to provide access to specific areas.

Common applications include lift motor rooms, access to large industrial equipment, large electrical pits, pump wells, container ports and transformers.



Note about weatherproofing

Due to the number of removable components including beams and individual covers, multi-part access covers by their very design cannot be made water tight. This is why *AS 3996: Access covers and grates* excludes large access cover systems from its scope. AS 3996 requires compliance in a number of areas including gas and water tightness. ACO multi-part covers are suitable for applications requiring general weather resistance on well drained enclosures. A reasonable degree of weather resistance can be achieved by sealing the edges of covers, frame interfaces and lifting holes each time the cover is removed for servicing. For applications requiring frequent access, ACO's Servokat Assist lift covers are recommended. See page 39.

Multi-part sizes

The table on page 23-25 provides information on the common multi-part sizes and is intended as a quick reference guide.

Multi-parts are available in numerous configurations and different cover sizes can be combined to create a clear opening to suit almost any requirement.

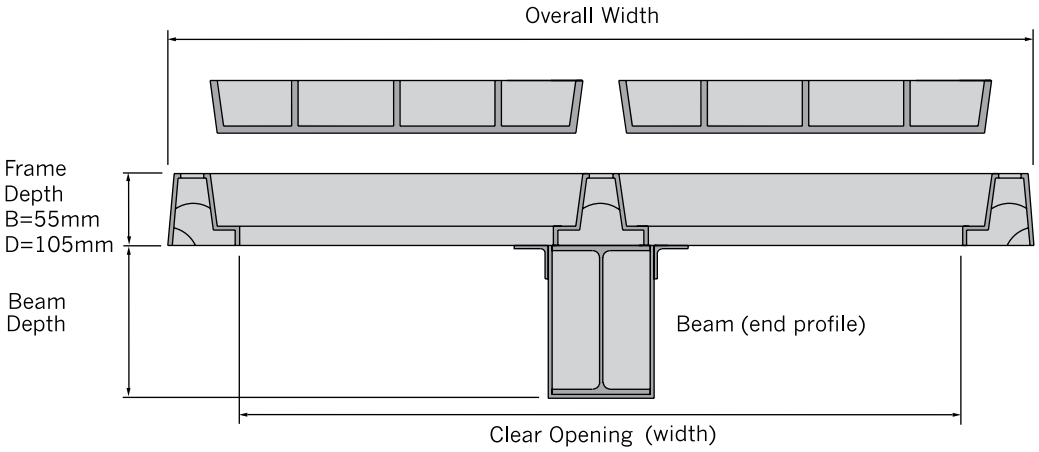
Multi-part design

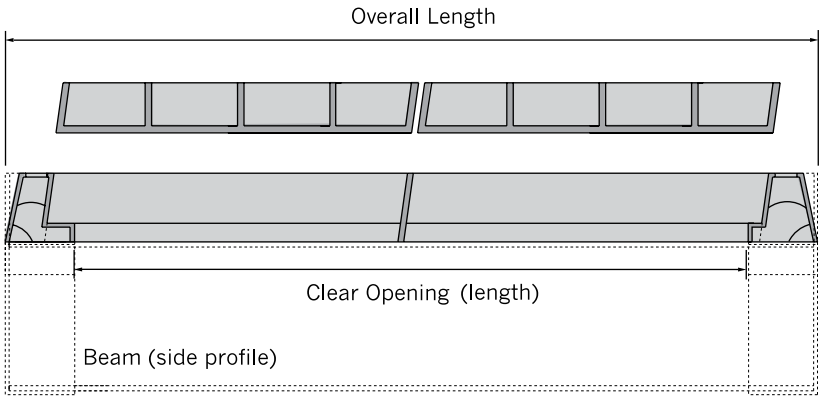
Due to the endless range of sizes available, ACO will design the multi-part closest to your requirements free of charge and obligation, simply supply the following information:

- Clear opening
- Load Class
- Position of obstacles that may affect cover removal direction
- Any width/slab depth restrictions

ACO will then provide:

- Actual clear opening
- Schedule of covers needed
- Rebate drawing (which is essential for correct installation)





Type	Part No.	Clear Opening* W x L (mm)	No Parts	Covers W x L	Load Bearing Beams Required	Decorative Edging*	Locking Option**
	Class B - AS 3996 - 80kN						
DIMP109B	**85817**	1000 x 945	4-Part	2 x 2	1 x 150mm UC	BE/SS - 12/40	LD/BB
DIMP1312B	**85822**	1300 x 1245	4-Part	2 x 2	1 x 150mm UC	BE/SS - 12/40	LD/BB
DIMP1323B	**85839**	1300 x 2340	6-Part	2 x 3	1 x 150mm UC	BE/SS - 12/40	LD/BB
DIMP1615B	**85843**	1600 x 1545	4-Part	2 x 2	1 x 150mm UC	BE/SS - 12/40	LD/BB
DIMP1925B	**85850**	1900 x 2535	8-Part	2 x 4	1 x 150mm UC	BE/SS - 12/40	LD/BB
DIMP2015B	**85864**	2000 x 1545	6-Part	3 x 2	2 x 150mm UC	BE/SS - 12/40	LD/BB
DIMP2019B	**85870**	2000 x 1935	12-Part	3 x 4	2 x 150mm UC	BE/SS - 12/40	LD/BB
DIMP2423B	**85888**	2450 x 2340	9-Part	3 x 3	2 x 150mm UC	BE/SS - 12/40	LD/BB
	Class D - AS 3996 - 210kN						
DIMP109D	**89819**	1050 x 970	4-Part	2 x 2	1 x 250mm UB	BE/SS - 12/40	LD/BB
DIMP1312D	**89823**	1350 x 1270	4-Part	2 x 2	1 x 250mm UB	BE/SS - 12/40	LD/BB
DIMP1323D	**89830**	1350 x 2390	6-Part	2 x 3	1 x 250mm UB	BE/SS - 12/40	LD/BB
DIMP1615D	**89844**	1650 x 1570	4-Part	2 x 2	1 x 250mm UB	BE/SS - 12/40	LD/BB
DIMP1926D	**89850**	1950 x 2610	8-Part	2 x 4	1 x 250mm UB	BE/SS - 12/40	LD/BB
DIMP2115D	**89868**	2100 x 1570	6-Part	3 x 2	2 x 250mm UB	BE/SS - 12/40	LD/BB
DIMP2120D	**89875**	2100 x 2090	9-Part	3 x 3	2 x 250mm UB	BE/SS - 12/40	LD/BB
DIMP2523D	**89882**	2550 x 2390	9-Part	3 x 3	2 x 250mm UB	BE/SS - 12/40	LD/BB

* Decorative Brass (BE) or Stainless Steel (SS) edging can be added - just specify BE or SS and height (12mm or 40mm) after part no.
** Optional locking bolts are available for security or back pressure applications - specify standard bolts (LD) or Barri bolts (BB) after part no.

Urbanfil® – Galvanised steel access covers and frames

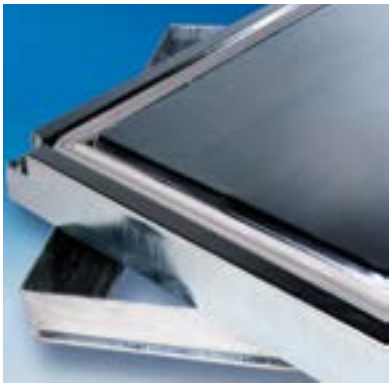
Typical Applications

- Internal areas
- Streetscapes
- Prestige/Aesthetic projects
- Light commercial areas
- Ducting trenches
- Custom size projects
- Electrical/communications
- Sport venues
- Bus/Rail platforms
- Shopping centres



Concrete ties provide an excellent fixing in surrounding concrete.





Seal provides a water and gas tight seal without the need for grease.

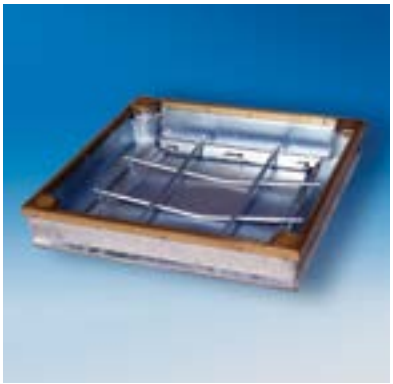


Four high tensile steel M16 recessed socketed locking bolts per cover with plastic cap to protect from dirt ingress.
Lifting bosses use standard AS 3996 lifting keys.

Standard cover



Decorative edge cover



Pavermate® – paver cover



Standard depth covers for use with tiles, concrete and epoxy floors; Pavermate® covers for use with maximum 40mm thick brick pavers. Custom sizes to order.

Product Features

- All covers are designed to meet Australian Standard AS 3996 Class B loadings
- 2mm thick steel frame and cover galvanised to AS 4680
- All covers have a polyethylene gas and water seal
- Seals odours
- Concrete ties on all frames to ensure good seating in surrounding concrete
- Four high tensile steel M16 locking bolts per cover, Barri bolts available for maximum security
- Lifting boss conforms to AS 3996 and uses Australian Standard lifting keys
- Steel reinforcing bars fixed to the base of all covers
- Frame and cover profile allows covers to be removed with a sliding action from all 4 sides
- Brass decorative edge option for superior aesthetics
- Stainless steel covers to special order, contact ACO for sizes and delivery



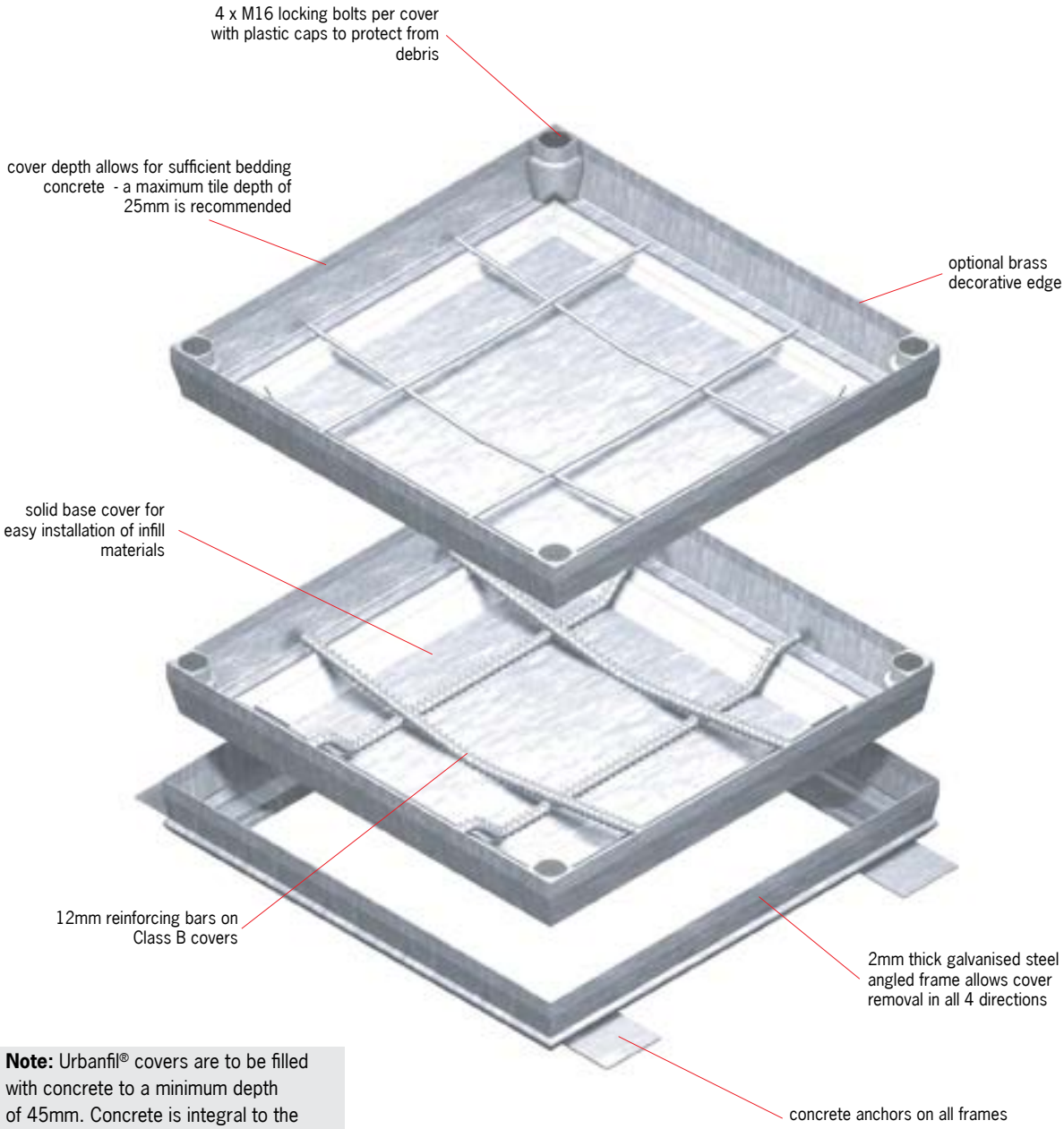


Urbanfil® – Single part galvanised steel recessed, lock and seal cover and frame

Single part units can be used to provide access to underground electricity and communication enclosures, tradewaste systems, valves and junction boxes.

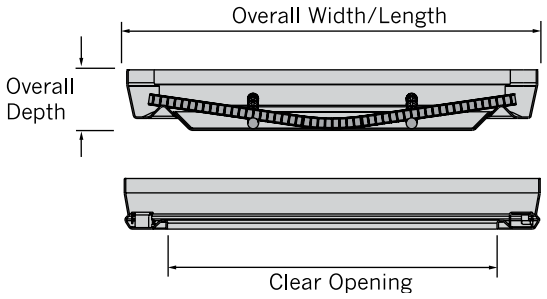
Urbanfil® covers should never be used in road applications.

For tiled finishes, tiles should be fully restrained and bonded to the concrete bed to prevent damage to cover. An epoxy mortar is recommended.



Note: Urbanfil® covers are to be filled with concrete to a minimum depth of 45mm. Concrete is integral to the strength of the cover. See page 41 for full details.

Class B



Type	Part No.	Clear Opening* W x L (mm)	Overall W x L (mm)	Overall Depth (mm)	Cover Wgt (kg)	Overall Wgt (kg)	Decorative Edging*	Locking Option**
	Class B - AS 3996 - 80kN							
GS33C	**81686**	300 x 300	430 x 430	85	7.1	11.9	BE	BB
GS44C	**81520**	450 x 450	580 x 580	85	10.7	16.8	BE	BB
GS46C	**81547**	450 x 600	580 x 730	85	13.2	19.9	BE	BB
GS66C	**81565**	600 x 600	730 x 730	85	15.7	23.0	BE	BB
GS67C	**81586**	600 x 750	730 x 880	85	18.6	26.7	BE	BB
GS69C	**82018**	600 x 900	730 x 1030	85	20.9	29.6	BE	BB
GS77C	**81601**	750 x 750	880 x 880	85	21.5	30.2	BE	BB
GS94C	**81661**	900 x 450	1030 x 580	85	17.6	25.7	BE	BB
GS99C	**81645**	900 x 900	1080 x 1080	85	27.6	37.5	BE	BB

* Decorative Brass (BE) edging can be added - just specify BE after part no.
** Optional security locking bolts are available for security applications - specify Barri bolts (BB) after part no.

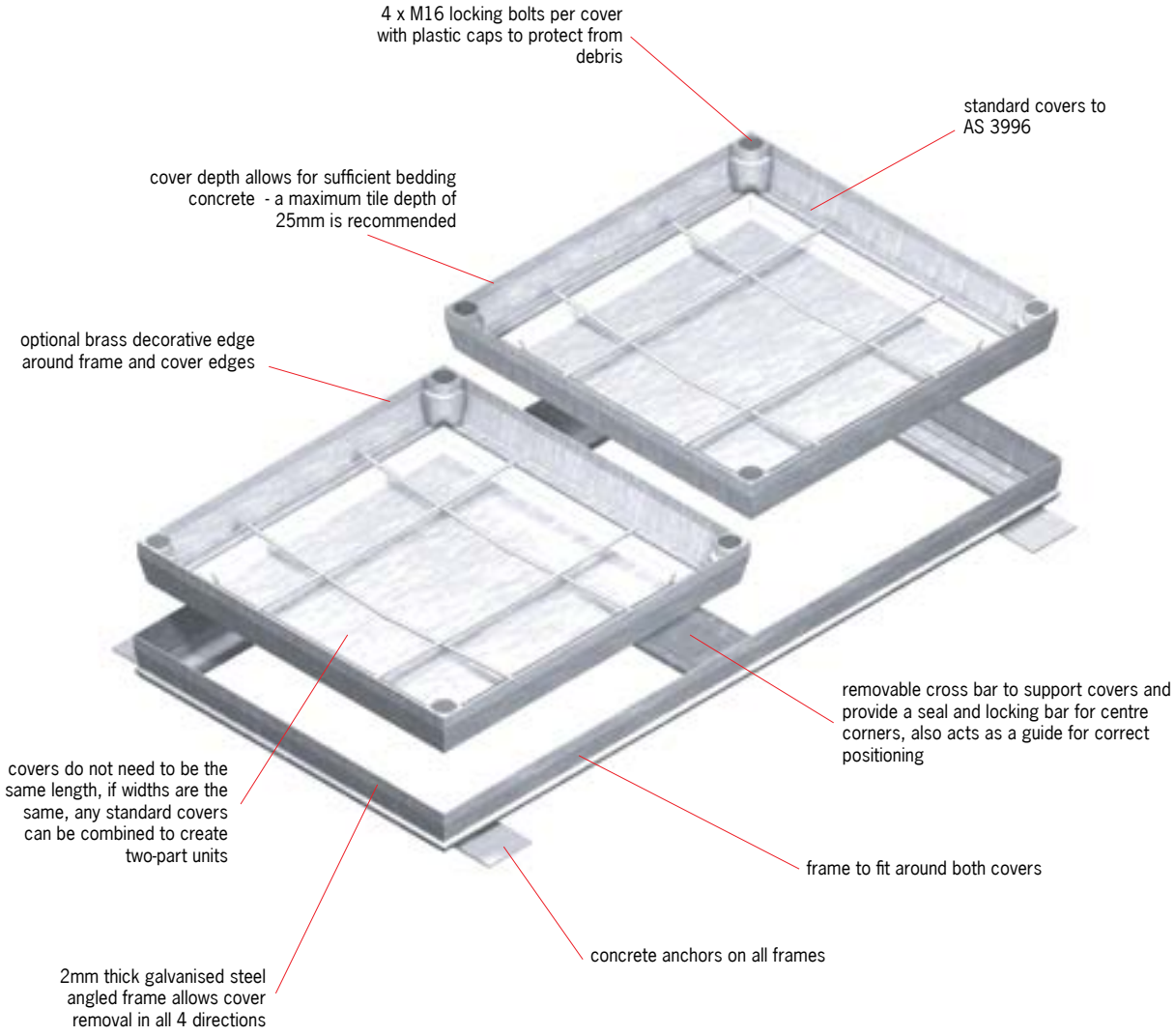
Urbanfil® – Two-part galvanised steel recessed, locking cover and frame

two-part units combine two standard size covers in a single run. They are used where a single part cover does not offer a large enough clear opening or if smaller individual covers are required for easier lifting.

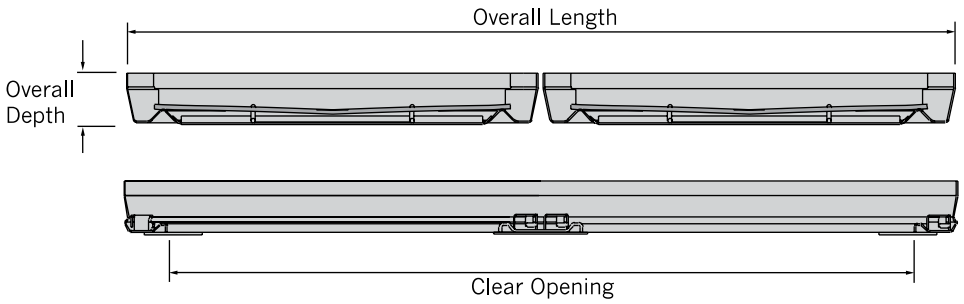
Common applications include grease traps/arrestors, electricity and communication enclosures, valves and junction boxes.

Urbanfil® covers should never be used in road applications.

For tiled finishes tiles should be fully restrained and bonded to the concrete bed to prevent damage to cover. An epoxy mortar is recommended.



Note: Urbanfil® covers are to be filled with concrete to a minimum depth of 45mm. Concrete is integral to the strength of the cover. See page 41 for full details.



Type	Part No.	Clear Opening* W x L (mm)	Overall W x L (mm)	Overall Depth (mm)	Cover Wgt (kg)	Overall Wgt (kg)	Decorative Edging*	Locking Option**
Class B - AS 3996 - 80kN								
GS2410C	**82521**	450 x 1020	580 x 1150	85	7.1/7.1	33.0	BE	BB
GS2411C	**82540**	450 x 1170	580 x 1300	85	7.1/10.7	36.1	BE	BB
GS2413C	**82567**	450 x 1320	580 x 1450	85	13.2/13.2	39.2	BE	BB
GS2610C	**82585**	600 x 1020	730 x 1150	85	13.2/13.2	39.2	BE	BB
GS2611C	**82606**	600 x 1170	730 x 1300	85	13.2/15.7	39.9	BE	BB
GS2613C	**82628**	600 x 1320	730 x 1450	85	15.7/15.7	45.5	BE	BB
GS2614C	**82640**	600 x 1470	730 x 1600	85	15.7/18.6	49.1	BE	BB
GS2616C	**82665**	600 x 1620	730 x 1750	85	18.6/18.6	52.0	BE	BB
GS2617C	**82681**	600 x 1770	730 x 1900	85	18.6/20.9	55.6	BE	BB
GS2619C	**82706**	600 x 1920	730 x 2050	85	20.9/20.9	58.6	BE	BB
GS2713C	**82724**	750 x 1320	880 x 1450	85	18.6/18.6	52.7	BE	BB
GS2714C	**82745**	750 x 1470	880 x 1600	85	18.6/21.6	56.3	BE	BB
GS2716C	**82762**	750 x 1620	880 x 1750	85	21.6/21.6	59.9	BE	BB
GS2913C	**82783**	900 x 1320	1030 x 1450	85	20.9/20.9	58.7	BE	BB

* Decorative Brass (BE) edging can be added - just specify BE after part no.
** Optional security locking bolts are available for security applications - specify Barri bolts (BB) after part no.

Pavermate® – Single part galvanised steel recessed, lock and seal cover and frame specifically for pavers

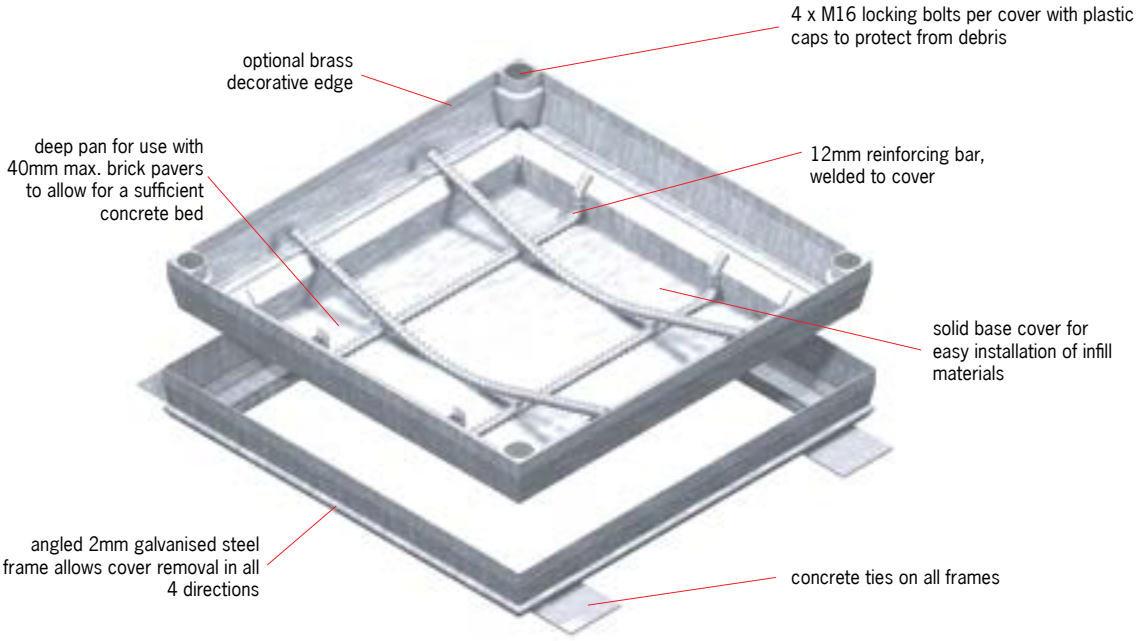
Single part units can be used to provide access to underground electricity and communication enclosures, valves and junction boxes.

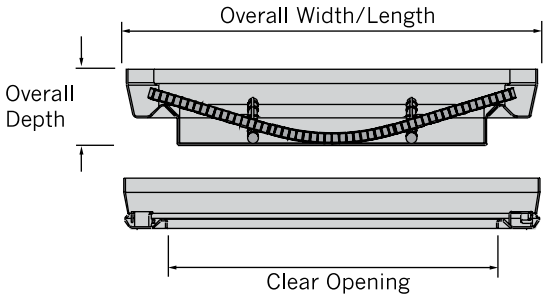
Pavermate® covers should never be used in road applications.

The deep pan profile allows for sufficient bedding concrete without having to cut pavers - making installation of brick pavers quick and simple.

Pavers should be fully restrained and bonded to the concrete bed to prevent damage to cover. An epoxy mortar is recommended.

The discreet edge detail is ideal for architectural solutions requiring visual continuity of the pavement.





Note: Pavermate® covers are to be filled with concrete to a minimum depth of 45mm. Concrete is integral to the strength of the cover. See page 41 for full details.

Type	Part No.	Clear Opening* W x L (mm)	Overall W x L (mm)	Overall Depth (mm)	Cover Wgt (kg)	Overall Wgt (kg)	Decorative Edging*	Locking Option**
	Class B - AS 3996 - 80kN							
PM33C	**81297**	300 x 300	430 x 430	135	6.7	1.2	BE	BB
PM44C	**81305**	450 x 450	580 x 580	135	11.4	17.0	BE	BB
PM46C	**81312**	450 x 600	580 x 730	135	12.5	18.9	BE	BB
PM66C	**81948**	600 x 600	730 x 730	135	14.9	21.9	BE	BB
PM67C	**81353**	600 x 750	730 x 880	135	17.8	25.4	BE	BB
PM69C	**81364**	600 x 900	730 x 1030	135	20.0	28.3	BE	BB
PM77C	**81372**	750 x 750	880 x 880	135	20.7	28.9	BE	BB
PM94C	**81338**	900 x 450	1030 x 580	135	17.9	25.5	BE	BB
PM99C	**81992**	900 x 900	1030 x 1030	135	26.5	36.0	BE	BB

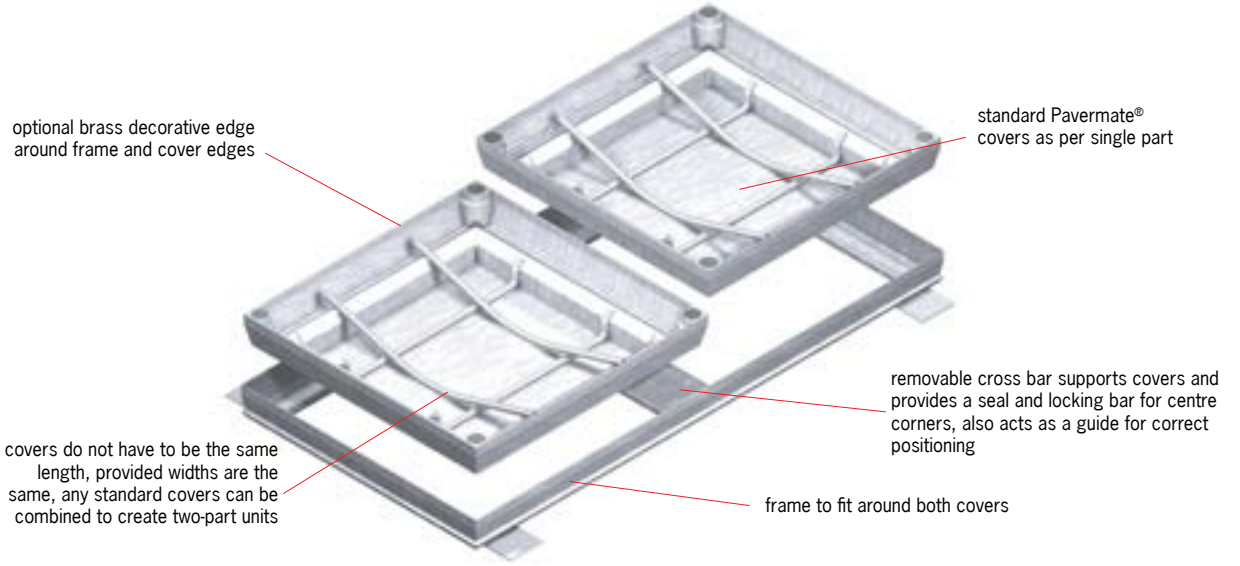
* Decorative Brass (BE) edging can be added - just specify BE after part no.
** Optional security locking bolts are available for security applications - specify Barri bolts (BB) after part no.

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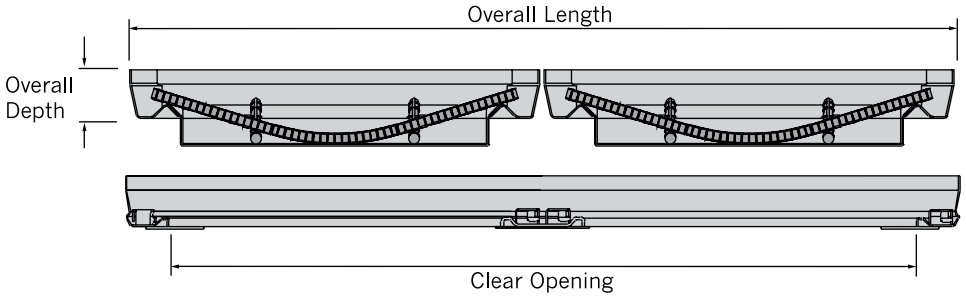
Pavermate® – Two-part galvanised steel recessed, locking cover and frame specifically for pavers

Two-part units combine two standard size covers in a single run. They are used where a single part cover does not offer a large enough clear opening or if smaller individual covers are required for easier lifting.

Common applications include grease traps/arrestors, electricity and communication enclosures, valves and junction boxes. Pavermate® covers should never be used in road applications.
The deep pan profile allows for sufficient bedding concrete without having to cut pavers - making installation of brick pavers quick and simple. Pavers should be fully restrained and bonded to the concrete bed to prevent damage to cover. An epoxy mortar is recommended.



Note: Pavermate® covers are to be filled with concrete to a minimum depth of 45mm. Concrete is integral to the strength of the cover. See page 41 for full details.



Type	Part No.	Clear Opening* W x L (mm)	Overall W x L (mm)	Overall Depth (mm)	Cover Wgt (kg)	Overall Wgt (kg)	Decorative Edging*	Locking Option**
	Class B - AS 3996 - 80kN							
PM2410B	**82358**	450 x 1020	580 x 1150	135	11.4/11.4	33.9	BE	BB
PM2411B	**82366**	450 x 1170	580 x 1300	135	11.4/12.5	35.9	BE	BB
PM2413B	**82373**	450 x 1320	580 x 1450	135	12.5/12.5	37.9	BE	BB
PM2610B	**82384**	600 x 1020	730 x 1150	135	12.5/12.5	39.7	BE	BB
PM2611B	**82398**	600 x 1170	730 x 1300	135	12.4/14.9	39.6	BE	BB
PM2613B	**82405**	600 x 1320	730 x 1450	135	14.9/14.9	39.2	BE	BB
PM2614B	**82417**	600 x 1470	730 x 1600	135	14.9/17.8	44.7	BE	BB
PM2616B	**82422**	600 x 1620	730 x 1750	135	17.8/17.8	50.4	BE	BB
PM2617B	**82436**	600 x 1770	730 x 1900	135	17.8/20.0	57.5	BE	BB
PM2619B	**82443**	600 x 1920	730 x 2050	135	20.0/20.0	56.9	BE	BB
PM2713B	**82450**	750 x 1320	880 x 1450	135	17.8/17.8	51.1	BE	BB
PM2714B	**82464**	750 x 1470	880 x 1600	135	17.8/20.7	54.7	BE	BB
PM2716B	**82471**	750 x 1620	880 x 1750	135	20.7/20.7	58.2	BE	BB
PM2913B	**82488**	900 x 1320	1030 x 1450	135	17.9/17.9	55.3	BE	BB

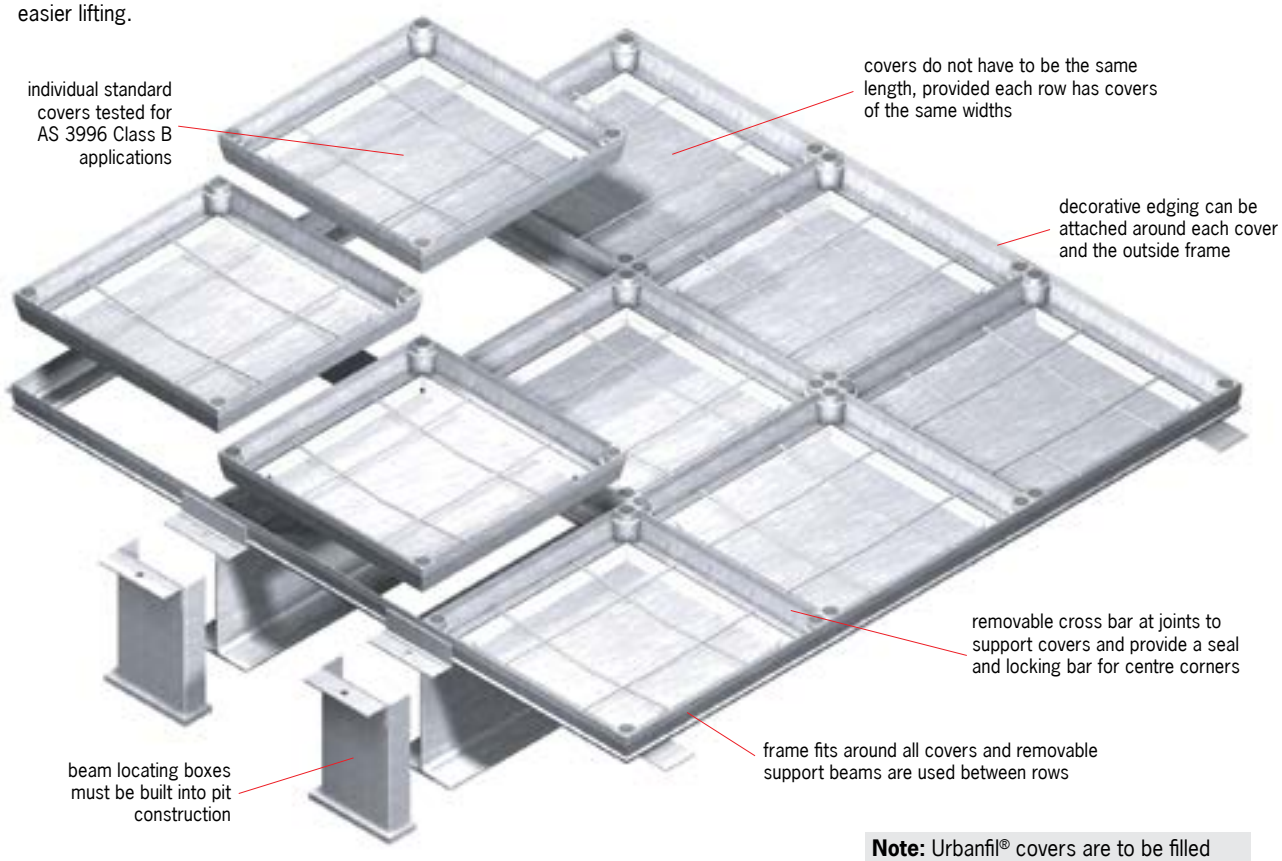
* Decorative Brass (BE) edging can be added - just specify BE after part no.
** Optional security locking bolts are available for security applications - specify Barri bolts (BB) after part no.

Urbanfil® – Multi-part galvanised steel recessed cover and frame

Multi-part units combine standard size covers in multiple rows and lengths. They are used where a single part cover does not offer a large enough clear opening or if smaller individual covers are required for easier lifting.

Common applications include lift motor rooms, access to large industrial equipment, pump wells and transformers.

All covers, cross bars and load bearing beams can be removed for full uninterrupted access or single covers can be removed to provide access to specific areas.



Note: Urbanfil® covers are to be filled with concrete to a minimum depth of 45mm. Concrete is integral to the strength of the cover. See page 41 for full details.

Note about weatherproofing

Due to the number of removable components including beams and individual covers, multi-part access covers by their very design cannot be made water tight. This is why *AS 3996: Access covers and grates* excludes large access cover systems from its scope. AS 3996 requires compliance in a number of areas including gas and water tightness. ACO multi-part covers are suitable for applications requiring general weather resistance on well drained enclosures. A reasonable degree of weather resistance can be achieved by sealing the edges of covers, frame interfaces and lifting holes each time the cover is removed for servicing. For applications requiring frequent access, ACO's Servokat Assist lift covers are recommended. See page 39.

Multi-part sizes

The table on page 33 provides information on common multi-part sizes and is intended as a quick reference guide.

Multi-parts are available in numerous configurations and different cover sizes can be combined to create a clear opening to suit almost any requirement.

Multi-part design

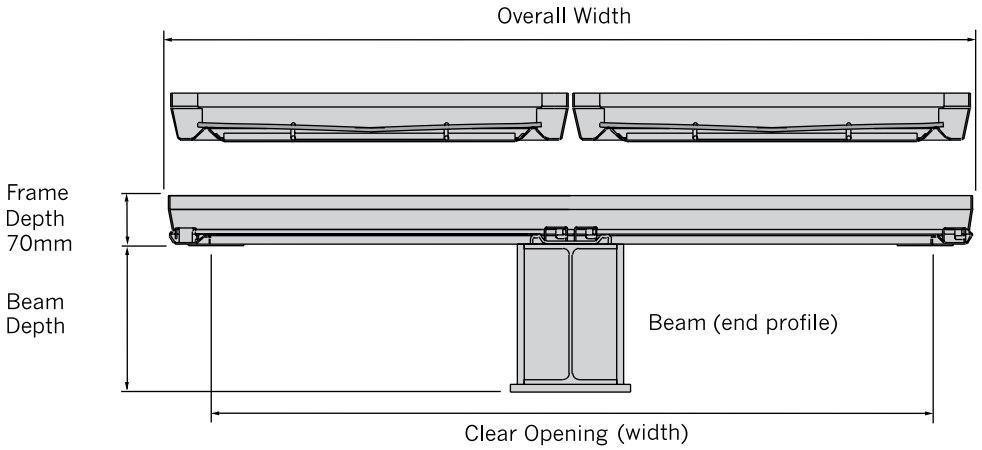
Due to the endless range of sizes available, ACO will design the multi-part closest to your requirements free of charge and obligation, simply supply the following information:

- Clear opening
- Load Class
- Any width/slab depth restrictions

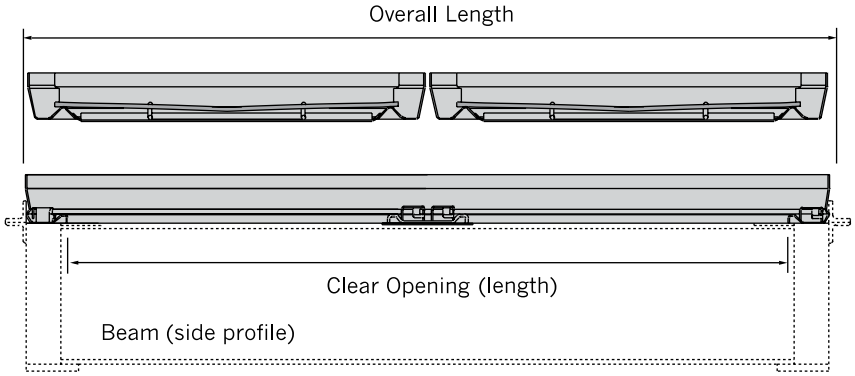
ACO will then provide:

- Actual clear opening
- Schedule of covers needed
- Rebate drawing (which is essential for correct installation)

Across the width of the Multi-Part



Across the length of the Multi-Part



Type	Part No.	Clear Opening* W x L (mm)	No Parts	Covers W x L	Load Bearing Beams Required	Decorative Edging*	Locking Option**
	Class B - AS 3996 - 80kN						
GSMP1010B	**83777**	1020 x 1020	4-Part	2 x 2	1 x 100mm UC	BE	BB
GSMP1313B	**83782**	1320 x 1320	4-Part	2 x 2	1 x 100mm UC	BE	BB
GSMP1515B	**83798**	1590 x 1590	9-Part	3 x 3	2 x 100mm UC	BE	BB
GSMP1521B	**83803**	1590 x 2160	12-Part	3 x 4	2 x 150mm UC	BE	BB
GSMP1624B	**83810**	1620 x 2490	6-Part	2 x 3	1 x 150mm UC	BE	BB
GSMP2020B	**83824**	2040 x 2040	9-Part	3 x 3	2 x 150mm UC	BE	BB
GSMP2424B	**83831**	2490 x 2490	9-Part	3 x 3	2 x 150mm UC	BE	BB

* Decorative Brass (BE) edging can be added - just specify BE after part no.
** Optional security locking bolts are available for security applications - specify Barri bolts (BB) after part no.

For other sizes please contact ACO

Saku – Thermoplastic access cover with Iron Frame

Typical Applications

- Sewer applications
- Stormwater
- Coastal areas
- Fuel and tradewaste
- Footpaths
- Residential yards



Work, Health & Safety

- One third the weight of traditional iron covers
- Minimises lifting injuries



Lifting provision

- Lifting hole with dust cap (removed) compliant to AS 3996



Strong reinforced ribs

- Meets AS 3996 Class B and prEN 124-6 Class B125
- Elevated temperature tested
- Sustained load tested
- Impact tested



Optional riser (make-up) rings

- Lightweight riser (make-up) rings (less than 5 kg each, for worker safety)
- Various depths and sloping variants available, to raise the cover for resurfacing
- Manufactured from HDPE defined in ASTM Specification D-4976

Circular cover



Product Features

- Lightweight thermoplastic cover - 8.3kg - one third the weight of traditional iron covers
- Appraised by the Water Services Association of Australia (WSAA)
- Covers are designed to Class B – AS 3996
- Gas and watertight to AS 3996
- Cover manufactured from polypropylene reinforced with glass fibres
- Ductile Iron frame with 8 bolt points for anchorage
- Optional HDPE riser (make-up) rings may be used to raise the access cover to finished surface levels
- Universal lifting key compatible
- Anti-slip Cover - rated to AS4586
- Low maintenance

WSAA Appraised [WSA PS - 293]





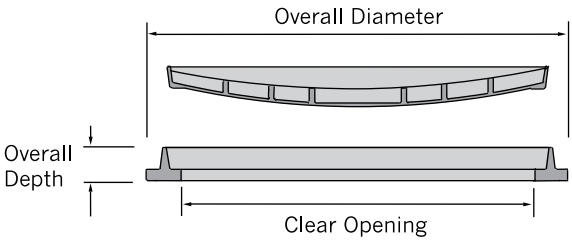
Saku – Circular solid top thermoplastic access cover and iron frame

ACO's SAKU cover and frame system is designed to comply with Australian urban industry requirements. The product consists of a lightweight thermoplastic cover and ductile iron frame with an EPDM rubber seal. Optional HDPE riser (make-up) rings may be used to raise the access cover to finished surface levels.

Manufactured from durable polypropylene, the cover is ideal for corrosive sewer environments, seaside applications and is gas and watertight to AS 3996.



Class B - 'T' Frame



Type	Part No.	Clear Opening Dia (mm)	Overall Dia (mm)	Overall Depth (mm)	Cover Wgt (kg)	Overall Wgt (kg)	Decorative edging[1]	Bolt options[5]
	Class B - AS 3996 - 80kN							
Cover and Frame								
Saku	**142026**	604	810	44	8.3	24.5	n/a	BB
Riser rings								
24R125	**142037**	-	852	32 [3]	-	1.9	n/a	n/a
24R150	**142031**	-	852	38 [3]	-	2.1	n/a	n/a
24R200	**142029**	-	852	51 [3]	-	2.5	n/a	n/a
24R400	**142030**	-	852	102 [3]	-	4.8	n/a	n/a
24S150	**142038**	-	852	38 [3 4]	-	1.8	n/a	n/a
Lifting Key	**84306**	-	-	-	-	0.6	n/a	n/a

1 Decorative edging - not available with Saku covers.
2 Dimensions of the frame (overall cover depth is 103mm)
3 Other depths are available by ordering different combinations of the 32mm, 38mm, 51mm and 102mm deep riser (make-up) rings
4 Sloped riser ring
5. Locking bolts are standard, for higher security specify Barri Bolts (BB) after Part No.

Servokat – Assisted lift access covers

Assisted lift covers are the ideal solution for applications where frequent access to the utilities or machinery beneath is required. The cover is hinged to the frame and a gas strut allows for easy opening.





Product Features

- Gas cylinder assisted lift on cover
- Rubber seal in frame ensures gas and water tight seal
- Cover can be opened from above or below
- A galvanised steel safety gate falls into place when the cover is lifted
- A stop latch holds the hinged cover open to prevent the cover accidentally closing



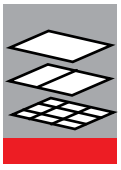
Loadings

- Covers are tested to EN 124 Class B (125kN), Class D (400kN) and Class F (900kN)
- EN124 uses a 250mm diameter test block



Sizes

- Typical, single part, clear opening sizes of 600 x 600, 1000 x 1000, 1200 x 1200, 1500 x 1500 are available as standard. Other sizes are also available
- A small selection of two-part covers are available (where one cover has assisted lift)



Styles

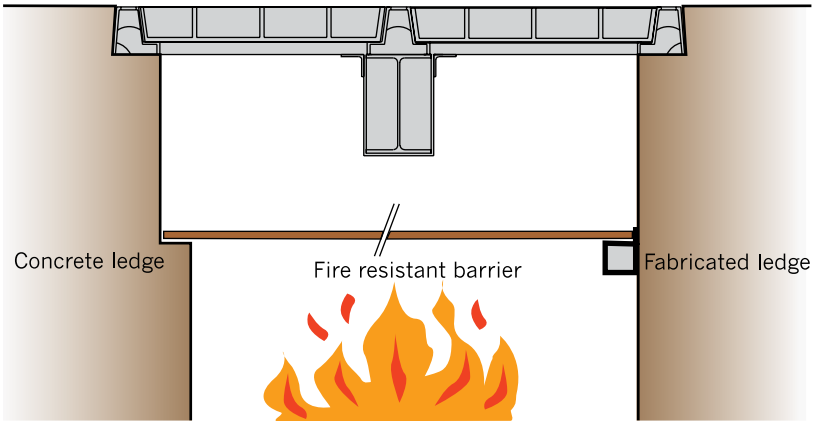
- Galvanised steel frame and solid top cover - loadings up to 400kN
- Galvanised steel frame and recessed cover (concrete filled) - loadings up to 900kN
- Cast iron frame and recessed cover - loadings up to 400kN
- Cast iron frame and recessed cover (concrete filled) - loadings up to 900kN

Contact ACO for full details

Fire rated covers

Certain applications may require that a cover offers a certain degree of fire resistance.

A fire rated cover is basically a standard access cover system which has a fire rated barrier installed beneath it to protect against flames, smoke, fumes and heat for up to 2 hours as required in AS 1530
Each system is custom made and ACO will work closely with you to supply a suitable fire rated system for the requirements of each application.
The following information is intended as a guide to the factors involved in specifying a fire cover.



Cover

- Any standard cover can be supplied as a fire rated cover
- Due to the addition of the fire barrier, clear opening will be reduced

Pit rebates

- The fire resistant barrier will need to be supported by either a formed ledge in the pit or a fabricated ledge - these ledges are used to support the framework which holds the fire resistant material (see diagram)

Safety gates

- Safety gates may be required in certain states to protect the non-load bearing fire rated barrier
- Safety gates fit to the sides of the pit below the cover
- A safety catch is required to hold the gates open to prevent them falling closed and trapping someone inside the pit

Fire resistance material

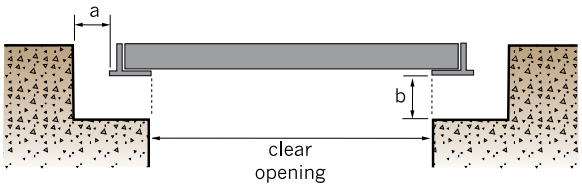
- An asbestos free material rated to 2 hours to AS 1530, higher ratings are also available

Installation and maintenance

Key Installation Considerations

- Load Class - select correct cover and rebate to suit load class. The frame is designed to be fully supported by a bed of concrete - see diagram below
- Install with cover positioned in the frame
- Decorative edging may require a deeper rebate than specified, depending on finish level
- Tiles or pavers should be fully restrained and bonded to the concrete bed. An epoxy mortar is recommended
- Cover and frame are a matching pair and should not be switched as fit and seal may be compromised
- Ensure that there is no rock between cover and frame otherwise seal will be compromised and service life will be reduced
- Concrete of a minimum strength of 32MPa and minimum aggregate of 12mm is recommended
- Position so that removal of cover is not obstructed by a wall, kerb etc.
- Applications where back pressure is expected may require additional reinforcing to hold the frame and cover in position. Engineering advice should be sought
- Urbanfil® / Pavermate® covers are to be filled to a minimum concrete depth of 45mm, with minimum strength 32MPa and minimum cement content of 400kg/m^3. Concrete is integral to the strength of the cover.

Installation Guide



Load Class	a	b
B	30	30
D	50	50
G	50	75

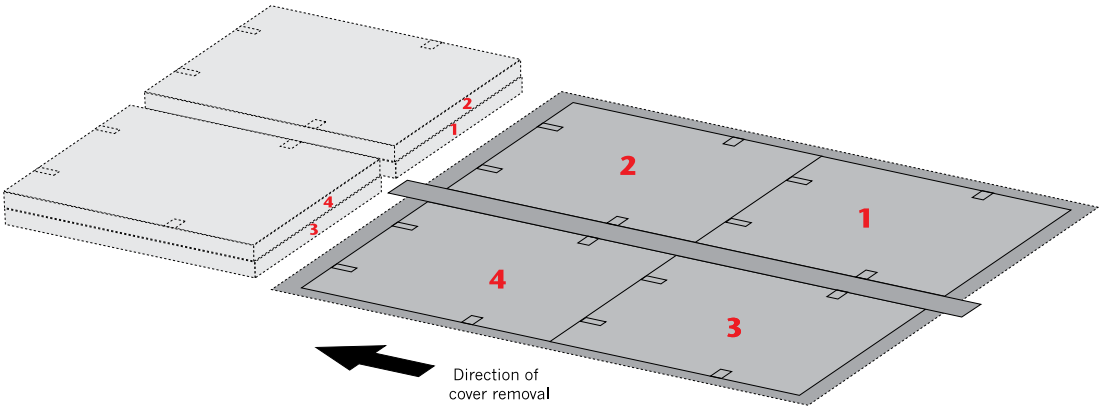
Rebate dimensions
Due to the varying frame dimension for different covers and decorative edging options, dimensions shown indicate the minimum amount of concrete required around the frame.

1. Form the access pit and rebate to size required - see diagram.
2. Position frame and cover in rebate.
3. Check unit is level and does not rock.
4. Pour concrete around frame and into cover (if applicable), vibrate to ensure all cavities in and around the frame are filled.
5. Level and finish concrete or lay pavers or pavement material.
6. Allow concrete to cure before removing cover - early removal may cause twisting of the frame.

Full details at **www.acoaccess.com.au**

Rhinocast® maintenance

To ensure long service and easy removal of covers, the following steps should be followed whenever lifting the cover or once annually whichever is sooner.



Tools Required:

- Lifting keys
- Shifting spanner
- 2 lengths of water pipe approx. 25mm dia (150mm longer than the width of the cover)
- Screwdriver
- Paint scraper
- Wire Brush
- Tin of grease

Note: square covers will fit diagonally through the frame so care should be taken not to drop the cover through the clear opening.

Rhinocast® Cover Replacement

1. Use a wire brush or scraper to remove any dirt, debris or rust from the cover and frame.
2. Liberally apply sealing grease to seat area of cover and frame.
3. Using long handled lifting keys and pipe sections to roll cover back into position and gently lower onto frame seat.
4. Lift and lower cover again to evenly distribute sealing grease.
5. For multiple covers replace in reverse order to ensure each cover is returned to its correct position. Replace dust caps.

Rhinocast® Cover Removal

1. Remove dust caps and ensure recesses are free from debris.
2. Insert key and rotate clockwise a quarter turn, lock in position. Protect decorative edging and pavement finish where necessary, position jacking screw over frame and screw down using shifting spanner to break the seal.

   If there are difficulties breaking seal, apply penetrating oil and allow a minimum of 24 hours, tap joint with hammer (taking care not to damage decorative edging) and try jacking screw again.
3. Once seal is broken, using standard, long handled or mobile lifting keys, lift the cover by leaning forward on the handles. Insert two sections of pipe to enable the cover to be rolled clear of the opening.
4. For multiple cover units remove covers in order and stack at the end of the row as shown on diagram above..

Rhinocast® Lifting Keys	
Short Handle (pair)	**84654**
Long Handle (pair)	**84854**

Short handle lifter - Part No. 84654



Long handle lifter - Part No. 84854



Urbanfil® maintenance

To ensure long service and easy removal of covers, the following steps should be followed whenever lifting the cover.

Tools Required:

- Short handle lifting key
- Screwdriver
- Locking key
- Brush

Note: square covers will fit diagonally through the frame so care should be taken not to drop the cover through the clear opening.

Urbanfil® Cover Replacement

1. Use soft brush to clean seat area of the frame.
2. Position covers in original position and orientation.
3. Replace bolts and dust caps.

Urbanfil® cover removal



Urbanfil® Cover Removal

1. Remove dust caps and use locking key to unbolt cover.
2. Use short handle lifting keys or mobile lifting equipment to remove cover(s).
3. For multiple cover units remove covers in order and stack at the end of the row as shown on diagram above.

Urbanfil® Lifting Keys	
Universal Lifting Key	**84306**
Locking Key	**84313**

Saku maintenance

To ensure long service and easy removal of covers, the following steps should be followed whenever lifting the cover or once annually whichever is sooner.

Tools Required:

- Lifting keys
- Brush



Universal Lifting Key
Part No. 84306

Saku Cover Replacement

1. Use a soft brush to ensure seal area is free from debris.
2. Reposition cover.
3. Replace bolts and dust caps.

Saku Cover Removal

1. Remove dust caps, use socket set to unbolt cover and use lifting key to remove cover.

Saku Lifting Key	
Universal Lifting Key	**84306**

Saku Chemical Resistance Table

Substance	Suitability
Acids – weak or dilute	Constant exposure
Acids – strong or concentrated	Intermittent
Alcohols, Aliphatic	Constant exposure
Aldehydes	Intermittent
Bases	Constant exposure
Esters	Intermittent
Aliphatic Hydrocarbons	Intermittent
Aromatic & Halogenated Hydrocarbons	Rare
Aromatic Ketones	Rare
Oxidising Agents	Occasional

The chemical resistance information contained in this table is indicative only and is based on reactions noted at ambient temperature. Higher temperatures will generally reduce the corrosion resistance of the materials.

Please contact ACO for further information regarding the suitability of a material.

Technical Support

ACO has an established Technical Services Department staffed with qualified engineers with many years experience advising on access covers.

This free service is offered with no obligation and is supported with high quality technical documentation.

Services offered include the following:

Multi-part scheduling and rebate drawings

Project specific documentation assists the installer prepare the rebate and position/remove the covers in the correct sequence.

ACO can provide a CAD technical drawing showing the requirements for the construction of the rebate prior to installation.

Allowances are made for placement of the side frames, end frames and the structural beams under the joists.

The CAD drawing has all the covers numbered to enable the installer to position and remove the covers in the correct sequence before the concrete pour and formwork stripping.

ACO needs the following information to produce the drawing:

- Pit clear opening
- Load class
- The position of obstacles that will restrict cover removal
- depth and width restrictions in the slab

ACCTRIX configurator for rhinocast multi-part system

To aid specification and pricing the ACCTRIX™ programme enables you to design your own multi-part system online:

http://www.acoaus.com.au/access/acctrix.php

http://www.acoaus.com.au/access/installation_rhinocast.htm

A choice of 5 systems based on your enclosure's size and load class is available. Once your preferences are made, a drawing is generated for specification purposes and submission to ACO for quotation.

ACO product lines

- **ACO Cablemate**
  A range of electrica and communication cable jointing pits and surface ducting systems.
- **ACO Drain**
  A range of grated trench drainage systems and pits made from polymer concrete. Grates are available in a large range of materials and finishes.
- **ACO Sport**
  A range of surface drainage systems and ancillary products for sports fields, running tracks and stadiums.
- **ACO Infrastructure**
  A range of trench drainage systems for roads, ports, airports and rail.
- **ACO StormBrixx**
  Attenuation, infiltration and detention stormwater management system.

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